E-COMMERCE SALES DASHBOARD

JANUARY 6, 2026

Table of Contents

1. Executive Summary 3
   1.1 Introduction 3
   1.2 Key Findings 4
   1.3 Conclusion and Recommendations 4
2. Body 5
   2.1 Data Source 5
      Data Origin 5
      Dataset Description 5
   2.2 Business Intelligence Questions 7
   2.3 Key Performance Indicators (KPIs) 7
   2.4 Findings Based on Analysis and Evaluation 8
   2.5 Conclusion and Recommendations 19
      Conclusion 19
      Recommendations 20
3 Appendix: BI Design 20
   3.1 Data Pre-Processing / Data Cleansing 20
      1. Dataset Loading 20
      2. Data Transformation Environment 21
      3. Column Renaming 22
      4. Data Type Standardization 22
      5. Data Validation and Quality Checks 24
      6. Handling of Null Values 26
      7. Conditional Formatting (Total Sales) 28
      viii. Product Price Binning 29
      ix. Creation of Conditional Column – Order Value Category 30
      x. Creation of Dimension Tables and Fact Table 31
   3.2 BI Data Modelling via Star Schema – Facts and Dimensions 36
      Before Star Schema Implementation 36
      After Star Schema Implementation 37
   3.3 DAX and M Language 39
   3.4 Dashboard 43
      Dashboard Structure 43

Page Organisation 43

1. Executive Summary

1.1 Introduction

The purpose of this report is to analyse an e-commerce dataset using **Microsoft Power BI** in order to identify key trends, patterns, and performance drivers related to sales, products, customers, and geographical markets. The analysis aims to support business decision-making by transforming transactional data into meaningful insights through interactive visualisations and calculated performance metrics.

The dashboard focuses on key performance indicators such as **Total Sales**, **Total Profit**, **Profit Margin**, and **Order Volume**, enabling stakeholders to assess overall business performance and identify areas for improvement. Selected visualisations from the full dashboard are included in this section to highlight the most significant findings.

Figure 1 Sales Trend Over Time

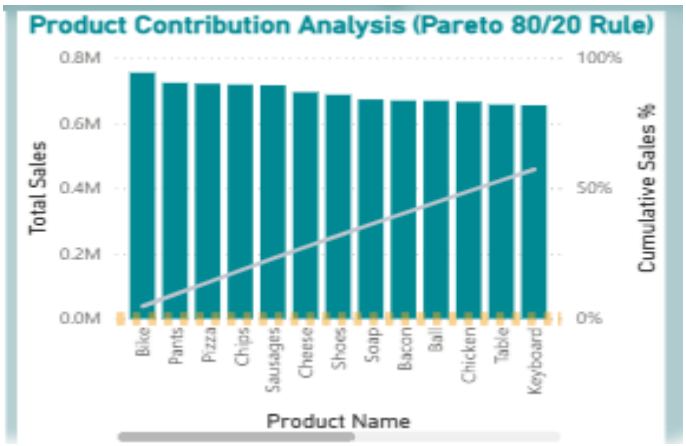


Figure 2 Product Contribution Analysis (Pareto Chart)

1.2 Key Findings

- Sales performance remains generally stable over the analysed period, with noticeable increases during specific timeframes, indicating the presence of seasonal demand patterns.
- Product contribution analysis shows that a small number of products generate a disproportionately large share of total revenue, confirming the applicability of the 80/20 rule.
- Customer analysis reveals that while most customers place low-to-medium value orders, a smaller group of high-value customers contributes significantly to overall revenue.
- Geographical analysis indicates uneven revenue distribution, with certain countries acting as primary revenue drivers while others underperform.
- Overall performance is driven by a limited set of products, customers, and regions, highlighting clear opportunities for targeted optimisation strategies.

1.3 Conclusion and Recommendations

In conclusion, the analysis highlights that business performance is driven by a limited number of products, customers, and geographical regions. While overall sales remain stable, there is significant potential to improve profitability through targeted strategic actions.

Based on the findings, the following recommendations are proposed:

- **Prioritise high-performing products** by ensuring optimal stock availability and increased marketing focus.
- **Develop targeted retention strategies** for high-value customers to maximise long-term revenue.
- **Leverage seasonal trends** by implementing data-driven promotional campaigns during peak periods.
- **Optimise regional strategies** by investing in high-performing markets while addressing challenges in underperforming regions.

The Power BI dashboard provides an effective decision-support tool by enabling stakeholders to explore these insights interactively and monitor performance in real time.

2. Body

This report analyses an e-commerce dataset using **Microsoft Power BI** to support business decision-making through data-driven insights. For the purpose of this analysis, the dataset is treated as originating from a real-world online retail business operating across multiple product categories and geographical regions. The business aims to better understand its sales performance, customer purchasing behaviour, product profitability, and regional demand patterns.

Like many e-commerce organisations, the business faces challenges such as identifying high-performing products, understanding customer value distribution, managing seasonal sales fluctuations, and optimising performance across different markets. Through interactive dashboards and analytical visualisations, this report seeks to address these challenges by transforming raw transactional data into meaningful insights that can guide strategic planning, marketing efforts, and operational improvements.

2.1 Data Source

The dataset used in this analysis is an **E-commerce transactional dataset**, provided in comma-separated values (CSV) format. The data represents historical sales transactions recorded by an online retail platform and includes information related to customers, products, orders, and geographical locations.

The dataset was imported into **Microsoft Power BI Desktop** and subsequently modelled using a star-schema structure to support efficient analysis.

Data Origin

- **Source:** The dataset E-commerce Dataset (CSV file) is from Kaggle.
- **Data Type:** Sample transactional sales data
- **Tool Used:** Microsoft Power BI Desktop

Dataset Description

Column Number	Column Name	Column Description	Column Data Type
1	Customer ID	Unique identifier assigned to each customer	Nominal

2	Customer Name	Name of the customer who placed the order	Nominal
3	Customer Email	Email address of the customer	Nominal
4	Customer Gender	Gender of the customer	Categorical
5	Country	Country where the customer is located	Nominal
6	Continent	Continent associated with the customer's country	Nominal
7	Product Name	Name of the product purchased	Nominal
8	Product Category	Category to which the product belongs	Categorical
9	Product Price	Unit price of the product	Numeric (Decimal)
10	Cost of Goods	Cost incurred to produce or acquire the product	Numeric (Decimal)
11	Quantity Sold	Number of units sold per transaction	Numeric (Integer)
12	Discount per Quantity	Discount applied per unit sold	Numeric (Decimal)
13	Sale Price	Total sale amount after discount	Numeric (Decimal)
14	Profit	Profit earned from the transaction	Numeric (Decimal)
15	Order Date	Date on which the order was placed	Date
16	Order Day	Day of the month when the order was placed	Numeric (Integer)
17	Order Month	Month in which the order was placed	Nominal

18	Order Year	Year in which the order was placed	Numeric (Integer)
19	Order Time	Time at which the order was placed	Time
20	Order Value Category	Categorisation of orders based on total value (Low, Medium, High)	Categorical

Table 1 Dataset Description

2.2 Business Intelligence Questions

1. **Is the business performing well in terms of overall sales, profit, and order value?**
2. How have sales and profit evolved over time, and are there signs of growth or volatility?
3. Which product categories and products contribute most to total revenue and profit?
4. Which products generate high sales volume versus high profit margins?
5. Are sales targets being met across product categories?
6. How is revenue distributed across customers and customer value segments?
7. Who are the top revenue-generating customers, and how concentrated is revenue?
8. How does purchasing behavior differ by customer gender?
9. Which regions and countries contribute most to total sales?
10. **How does average order value relate to profitability across product categories, and which categories offer the best balance between revenue per order and profit margin?**

2.3 Key Performance Indicators (KPIs)

Key Performance Indicators (KPIs) were defined to measure critical business processes related to sales performance, profitability, product effectiveness, and customer behaviour. These KPIs support performance monitoring and provide the foundation for all analytical visuals within the dashboard.

- **Total Sales:** Calculates the total revenue generated from all sales transactions.
- **Total Profit:** Measures the total profit earned across all transactions.
- **Total Quantity Sold:** Represents the total number of units sold.
- **Profit Margin (%):** Indicates profitability by measuring profit as a proportion of total sales.
- **Average Order Value (AOV):** Calculates the average value of customer orders based on total sales.

- **Cumulative Sales (%):** Measures the cumulative contribution of products to total sales, supporting Pareto (80/20) analysis.
- **Loss Orders:** Counts the number of orders that resulted in a loss (negative profit).
- **Customer Rank:** Ranks customers based on total sales contribution.
- **Average Discount:** Calculates the average discount applied per quantity sold.
- **Discounted Orders:** Measures the number of orders where a discount was applied.

2.4 Findings Based on Analysis and Evaluation

1. **Is the business performing well in terms of overall sales, profit, and order value?**

- **Visuals Used:** KPI Cards (Total Sales, Total Profit, Profit Margin %, Average Order Value)
- **Data & KPIs Used:** Total Sales, Total Profit, Profit Margin %, Average Order Value
- **Why This Visual:** KPI cards provide an immediate high-level summary of business performance and are suitable for executive-level evaluation.
- **Key Findings:** The business demonstrates strong financial performance, with high total sales and profit and a profit margin of approximately 45%, indicating effective pricing and cost management.

Figure 3 KPIs

2. How have sales and profit evolved over time, and are there signs of growth or volatility?

Charts Used

- Revenue Trend Over Time (Line Chart)
- Sales Composition Trend (Area Chart)
- Monthly Sales Pattern Across Years (Stacked column chart)

Data Used

- Order dates (year, month)
- Total sales

- Total profit

Why This Chart

Line and area charts are ideal for trend analysis and seasonality detection across multiple periods.

Key Insights

- Sales show **overall growth with minor fluctuations**, peaking mid-period.
- Certain years exhibit stronger growth momentum, while others indicate plateauing.
- Monthly patterns suggest **seasonal spikes**, especially toward later months.

Figure 4 Revenue Trend Over Time

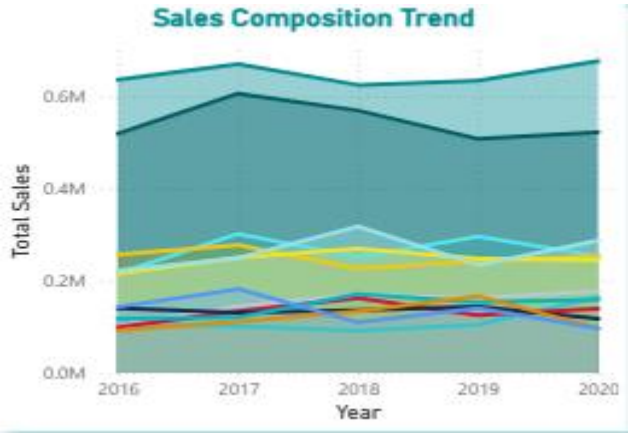


Figure 5 Area Chart

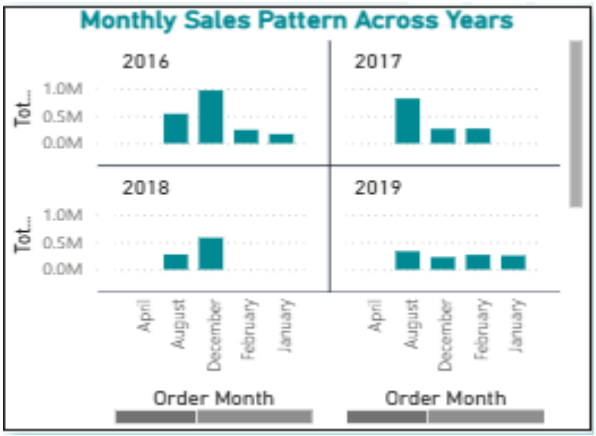


Figure 6 Stacked column chart with small multiples

3. Which product categories and products contribute most to total revenue and profit?

Charts Used

- Profit Contribution by Category (Waterfall Chart)
- Product Category Sales Rank Dynamics Over Time (Ribbon Chart)
- Product Contribution Analysis (Pareto Chart)

Data Used

- Product category
- Product-level sales
- Profit

Why This Chart

Waterfall charts explain cumulative contribution, while Pareto analysis highlights revenue concentration. The ribbon chart serves the purpose of trend analysis over time.

Key Insights

- A small number of categories drive the majority of revenue and profit.
- Certain categories consistently dominate across years, indicating stable demand.
- The Pareto analysis confirms that **~20% of products generate a disproportionate share of total sales**.

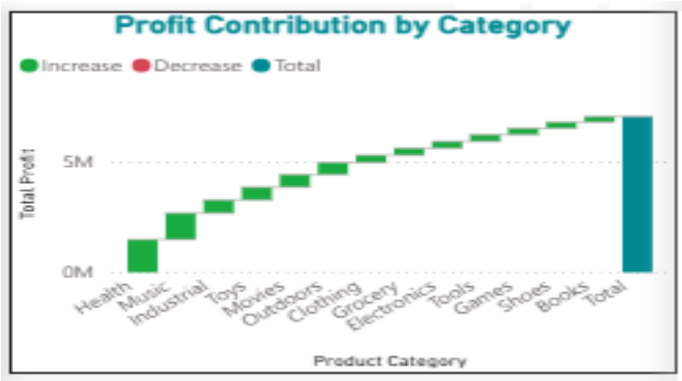


Figure 7 WaterFall chart

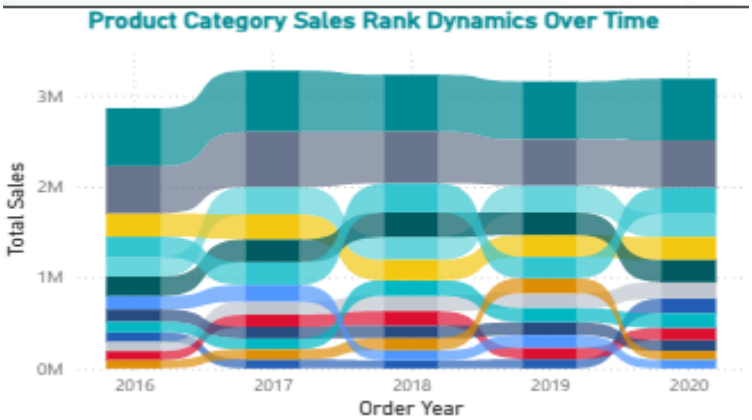


Figure 8 Ribbon Chart

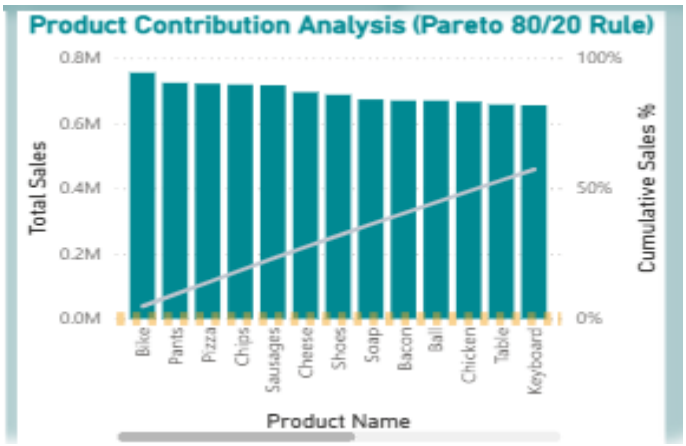


Figure 9 Product Contribution Analysis

4. Which products generate high sales volume versus high profit margins?

Charts Used

- Product Profitability vs Order Value (Scatter Plot)
- Product Price Distribution (Treemap)

Data Used

- Average order value
- Profit margin
- Product sales

Why This Chart: Scatter plots clearly show trade-offs between profitability and order value, while treemaps visualize product contribution.

Key Insights

- Some high-volume products operate on **lower margins**, indicating potential pricing optimization opportunities.
- Several niche products deliver **high margins despite lower sales volume**, making them strategic profitability drivers.

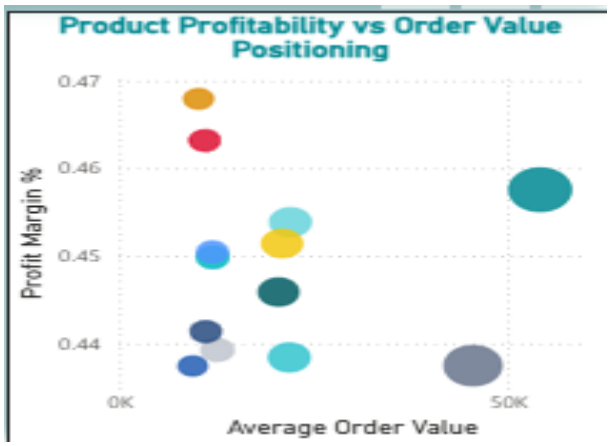


Figure 10 Scatter Plot

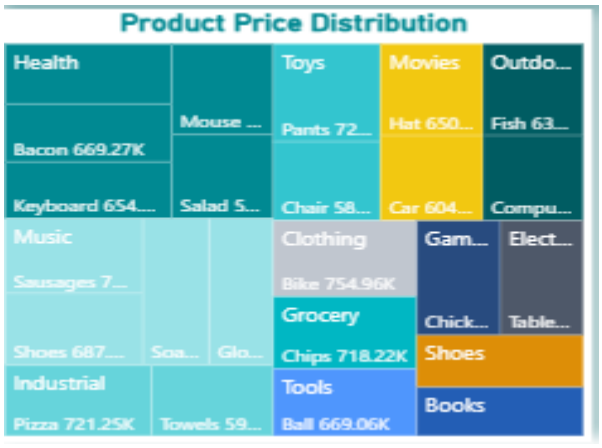


Figure 11 Treemap

5. Are sales targets being met across product categories?

Charts Used

- Total Sales Actual vs Target (Bullet Chart)
- Total Sales vs Target (Gauge Chart)

Data Used

- Target sales
- Actual sales
- Product category

Why This Chart

Bullet charts are ideal for comparing performance against predefined benchmarks. Gauge chart analyze if the target is achieved or not.

Key Insights

- Several categories meet or exceed targets, indicating strong execution.
- Underperforming categories highlight areas requiring corrective action or reassessment of targets.

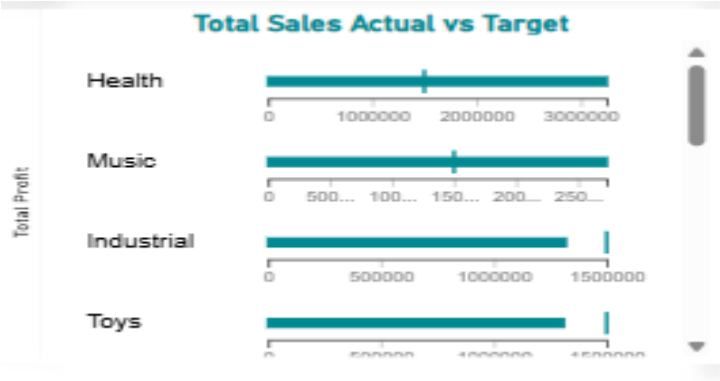


Figure 12 Bullet Chart

Figure 13 Gauge Chart

6. How is revenue distributed across customers and customer value segments?

Charts Used

- Customer Spend Distribution (Histogram)

- Customer Value Segmentation Funnel

Data Used

- Customer sales
- Order value
- Customer segmentation logic

Why This Chart

Histograms show spend spread, while funnels explain customer concentration across value tiers.

Key Insights

- Revenue is **heavily concentrated among high-value customers**.
- Low-value customers form a large base but contribute relatively less to total revenue.
- Retaining and upselling high-value customers is critical.

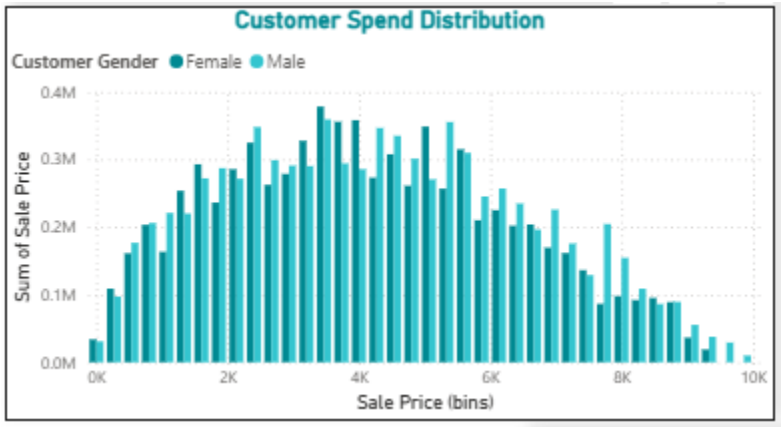


Figure 14 Histogram

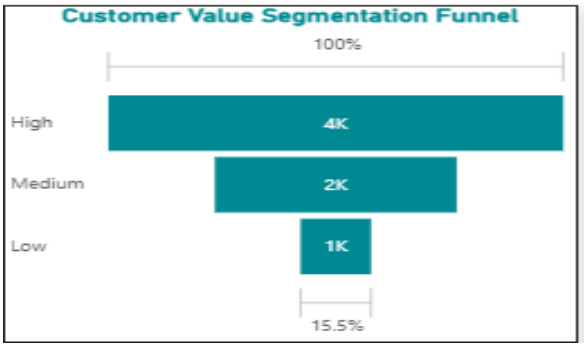


Figure 15 Funnel Chart

7. Who are the top revenue-generating customers, and how concentrated is revenue?

Charts Used

- Top Revenue Contributors Table

Data Used

- Customer name
- Total orders
- Total sales
- Profit margin

Why This Chart

Tables allow precise comparison and ranking of individual customer performance.

Key Insights

- A small group of customers contributes a significant portion of total revenue.
- High-performing customers also show strong profit margins, making them strategically important.

Top Revenue Contributors

Customer Name	Total Orders	Total Sales	Profit Margin %
Deshawn Marquardt	4	28,089.39	0.58
Trystan Nolan	6	25,314.30	0.49
Lorenz Hessel	4	23,736.39	0.47
Nayeli Walter	5	23,316.64	0.49
Helga Kiehn	4	22,629.66	0.27
Keegan Bernhard	6	21,988.45	0.62
Kiara Vandervort	4	21,313.80	0.41
London Ortiz	5	20,928.36	0.51
Matteo Block	5	20,849.24	0.38
Elnora Harvey	3	20,844.48	0.53
Jude Dietrich	4	20,832.91	0.38
Imogene Pacocha	3	20,666.70	0.39
Total	**6104**	**15,717,504.19**	**0.45**

Figure 16 Top Revenue Contributors Table

8. How does purchasing behavior differ by customer gender?

Charts Used

- Revenue Contribution by Gender (Donut Chart)

- Average Order Value Distribution by Gender (Violin Plot)

Data Used

- Customer gender
- Sales
- Order value

Why This Chart

Donut charts show proportional contribution, while violin plots reveal distribution patterns.

Key Insights

- Revenue contribution is relatively balanced across genders.
- One gender exhibits a **higher average order value**, indicating differentiated purchasing behavior.

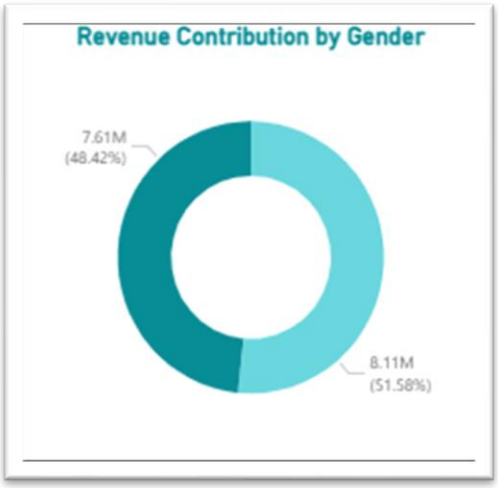


Figure 17 Donut Chart

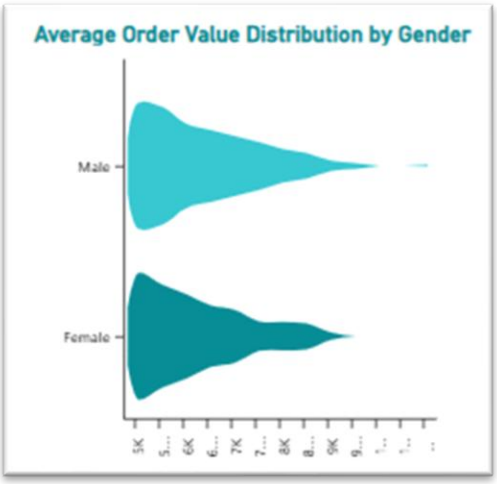


Figure 18 Violin Chart

9. Which regions and countries contribute most to total sales?

Charts Used

- Regional Sale Price Map
- Sales Drivers by Country (Decomposition Tree)

Data Used

- Country
- Region
- Sales value

Why This Chart

Maps provide geographic context, while hierarchical drivers explain contribution depth.

Key Insights

- Certain regions consistently outperform others.
- Some smaller markets show strong sales potential despite lower visibility.

Figure 19 Regional Sales

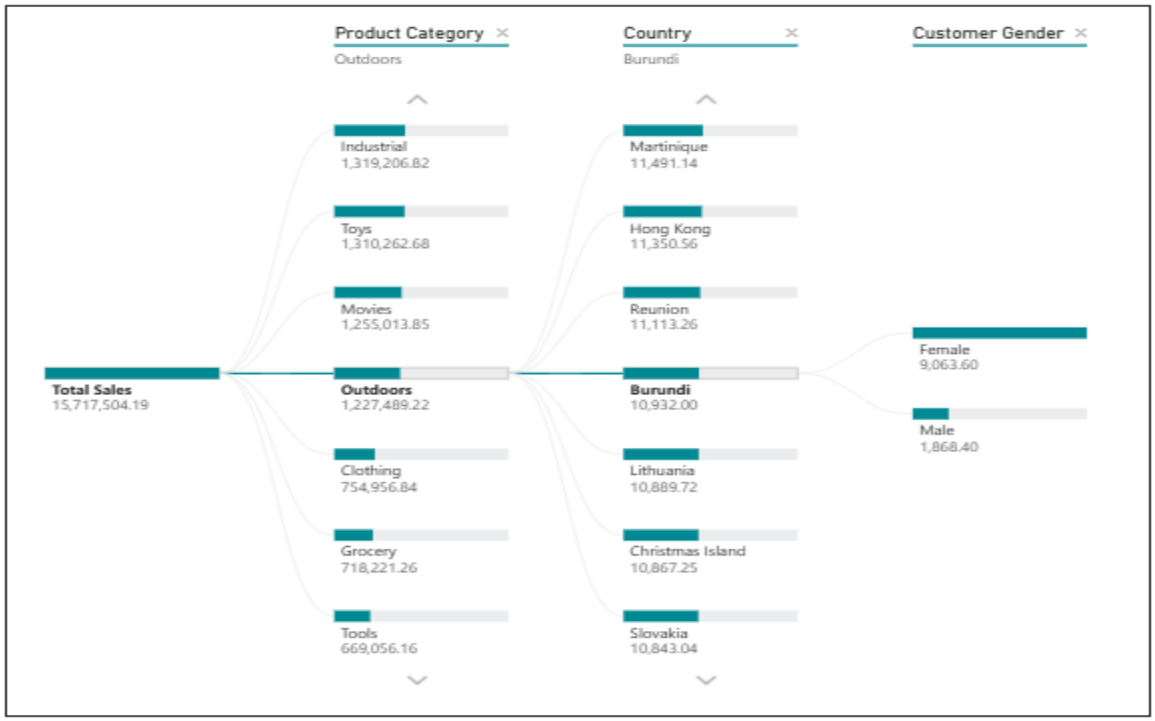


Figure 20 Decomposition Tree

10. **How does average order value relate to profitability across product categories, and which categories offer the best balance between revenue per order and profit margin?**

Charts Used

- Order Value vs Profitability by Category (Scatter Plot)

Data Used

- Product category
- Average order value
- Profit margin percentage

Why This Chart

A scatter plot is ideal for comparing **two quantitative measures simultaneously** (order value vs profit margin), Identifying **trade-offs**, clusters, and outliers and highlighting categories that achieve **both high order value and high profitability.**

Key Insights

- Some categories generate **high average order values but operate on lower margins**, indicating pricing or cost-structure challenges.
- Other categories maintain **strong profit margins despite moderate order values**, making them reliable profitability drivers.
- Categories positioned in the **upper-right quadrant** represent the most strategically valuable segments, combining high revenue per order with strong margins.
- The visualization supports **portfolio optimization decisions**, such as which categories to scale, optimize, or reassess.

Figure 21 Order Value vs Profitability by Category (Scatter Plot)

2.5 Conclusion and Recommendations

Conclusion

The analysis provides a holistic view of business performance across **sales, profit, customers, products, regions, and sales drivers**. Overall performance is strong, with healthy sales, stable profit margins, and a solid average order value. Sales trends show general growth with some volatility, indicating opportunities for better forecasting and planning.

Revenue and profit are **concentrated among a limited set of products, categories, and high-value customers**, confirming dependency risks but also clear optimization opportunities. Product analysis highlights a trade-off between high-volume and high-margin categories, while customer analysis shows that a small segment contributes most of the revenue. Regional and sales driver insights reveal that

specific combinations of product, geography, and customer demographics drive overall performance. Target analysis identifies performance gaps in certain categories.

Recommendations

i. **Prioritize high-value customers** through retention and loyalty initiatives to protect core revenue streams.

ii. **Optimize the product portfolio** by scaling high-margin categories and reviewing low-margin, high-volume products.

iii. **Refine pricing strategies** where high order value does not translate into strong profitability.

iv. **Reassess sales targets** for underperforming categories and strengthen performance monitoring.

v. **Focus on high-performing regions** while exploring growth opportunities in underpenetrated markets.

vi. **Align marketing efforts** with customer behavior and purchasing patterns to increase order value.

vii. **Continuously monitor sales drivers** to support informed strategic and operational decisions.

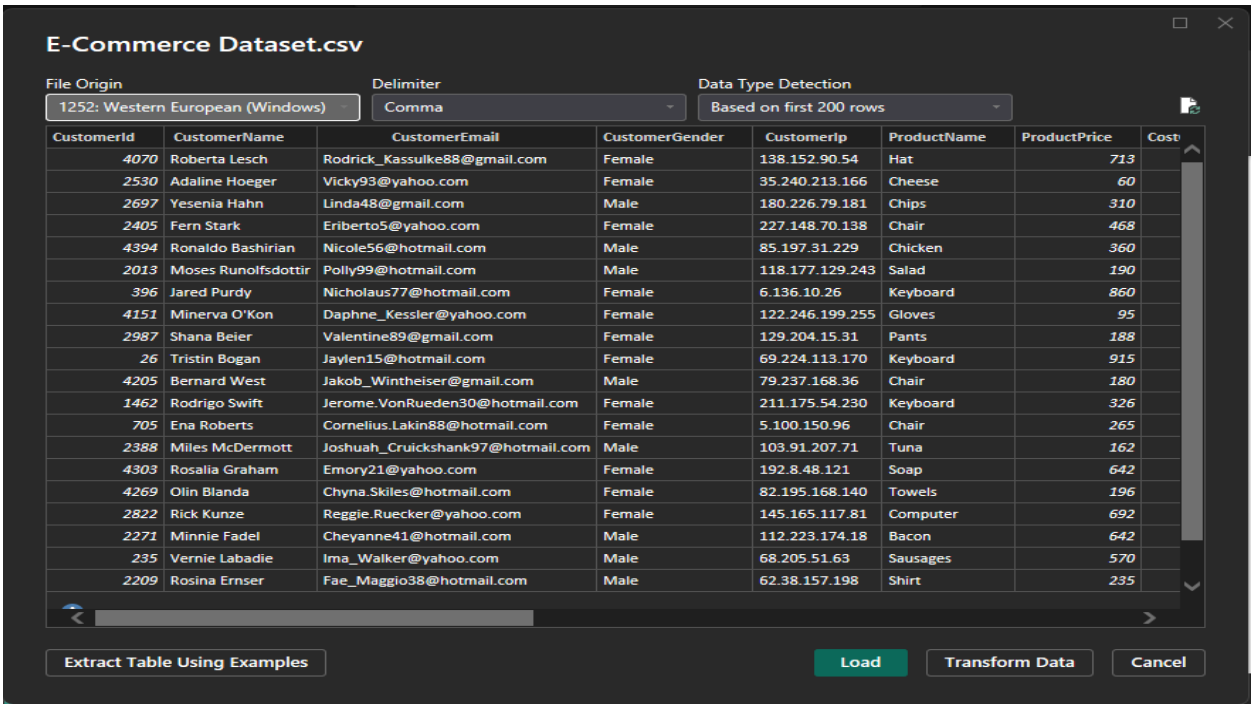
3 Appendix: BI Design

3.1 Data Pre-Processing / Data Cleansing

This section describes the data pre-processing and cleansing steps performed in **Power BI (Power Query Editor)** to prepare the E-Commerce dataset for analysis and visualization. The objective of these steps was to ensure **data accuracy, consistency, usability, and analytical readiness**.

1. Dataset Loading

- The raw dataset (E-Commerce Dataset.csv) was imported into Power BI using the **Text/CSV connector**.
- The correct **file origin (UTF-8 / Western European encoding)** and **comma delimiter** were applied to ensure proper character interpretation.
- The dataset was successfully loaded into Power Query for transformation.



CustomerId	CustomerName	CustomerEmail	CustomerGender	CustomerIp	ProductName	ProductPrice	Cost
4070	Roberta Lesch	Rodrick_Kassulke88@gmail.com	Female	138.152.90.54	Hat	713	
2530	Adaline Hoeger	Vicky93@yahoo.com	Female	35.240.213.166	Cheese	60	
2697	Yesenia Hahn	Linda48@gmail.com	Male	180.226.79.181	Chips	310	
2405	Fern Stark	Eriberto5@yahoo.com	Female	227.148.70.138	Chair	468	
4394	Ronaldo Bashirian	Nicole56@hotmail.com	Male	85.197.31.229	Chicken	360	
2013	Moses Runolfsdottir	Polly99@hotmail.com	Male	118.177.129.243	Salad	190	
396	Jared Purdy	Nicholaus77@hotmail.com	Female	6.136.10.26	Keyboard	860	
4151	Minerva O'Kon	Daphne_Kessler@yahoo.com	Female	122.246.199.255	Gloves	95	
2987	Shana Beier	Valentine89@gmail.com	Female	129.204.15.31	Pants	188	
26	Tristin Bogan	Jaylen15@hotmail.com	Female	69.224.113.170	Keyboard	915	
4205	Bernard West	Jakob_Wintheiser@gmail.com	Male	79.237.168.36	Chair	180	
1462	Rodrigo Swift	Jerome.VonRueden30@hotmail.com	Female	211.175.54.230	Keyboard	326	
705	Ena Roberts	Cornelius.Lakin88@hotmail.com	Female	5.100.150.96	Chair	265	
2388	Miles McDermott	Joshuah_Cruickshank97@hotmail.com	Male	103.91.207.71	Tuna	162	
4303	Rosalia Graham	Emory21@yahoo.com	Female	192.8.48.121	Soap	642	
4269	Olin Blanda	Chyna.Skiles@hotmail.com	Female	82.195.168.140	Towels	196	
2822	Rick Kunze	Reggie.Ruecker@yahoo.com	Female	145.165.117.81	Computer	692	
2271	Minnie Fadel	Cheyanne41@hotmail.com	Male	112.223.174.18	Bacon	642	
235	Vernie Labadie	Ima_Walker@yahoo.com	Male	68.205.51.63	Sausages	570	
2209	Rosina Ernser	Fae_Maggio38@hotmail.com	Male	62.38.157.198	Shirt	235	

Figure 22 Dataset Loading

2. Data Transformation Environment

- The **Power Query Editor** was used as the primary environment for data cleaning and preparation.
- Column profiling was enabled to inspect **valid, error, and empty values** across fields.

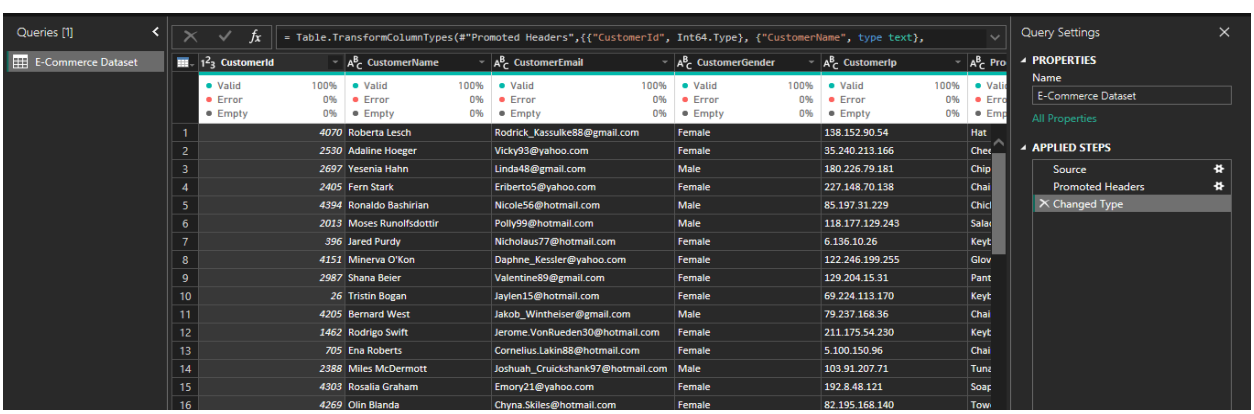


Figure 23 Data Transformation Environment

3. Column Renaming

- Column headers were renamed to improve **clarity, consistency, and readability**.
- Naming conventions were standardized (e.g., CustomerId → Customer ID, ProductPrice → Product Price, CostOfGoods → Cost Of Goods).
- Spaces and descriptive naming were applied to align with BI reporting standards.

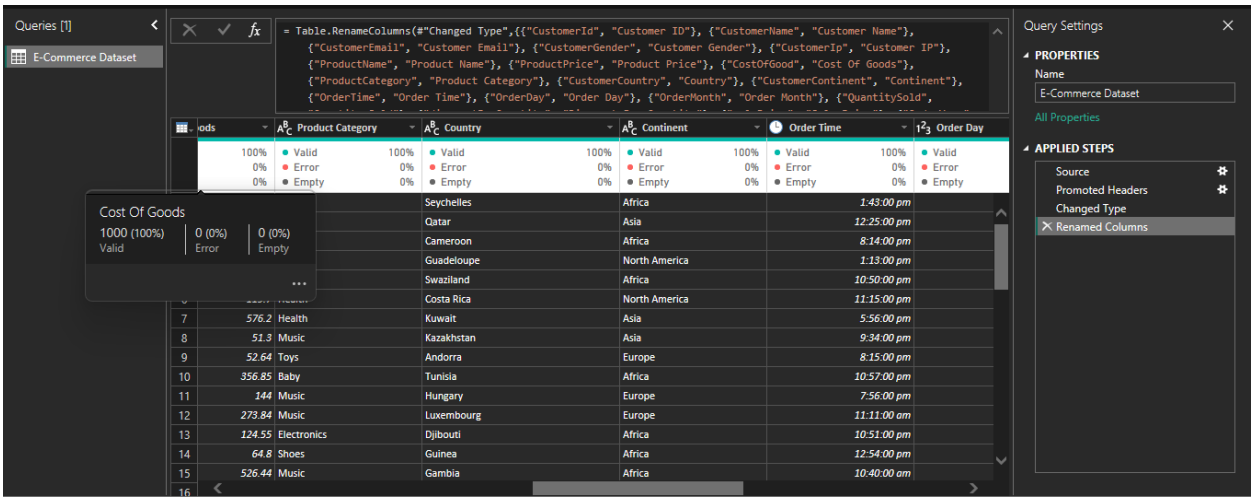


Figure 24 Column Renaming

4. Data Type Standardization

- Data types were explicitly set to prevent incorrect aggregations and calculation errors:
  - Numeric fields (e.g., Product Price, Cost of Goods, Quantity Sold, Discount) → **Number**
  - Date and time fields → **Date / Time**
  - Categorical fields (e.g., Gender, Country, Product Category) → **Text**
- Incorrect or ambiguous auto-detected data types were corrected manually.

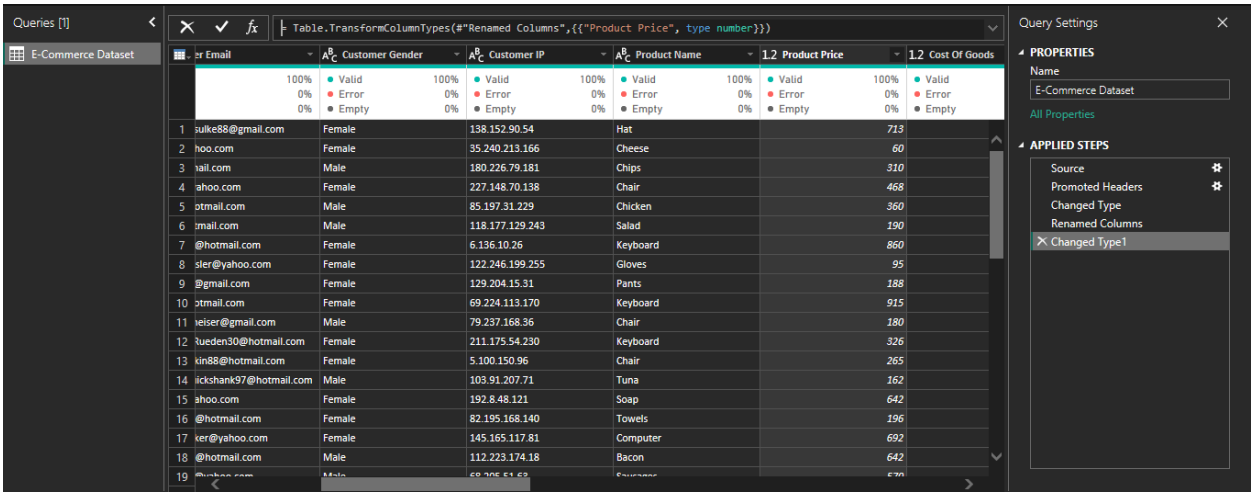


Figure 25 Data Type Standardization

- Changed the data type from Numerical to Date/Time in Order Date custom column.

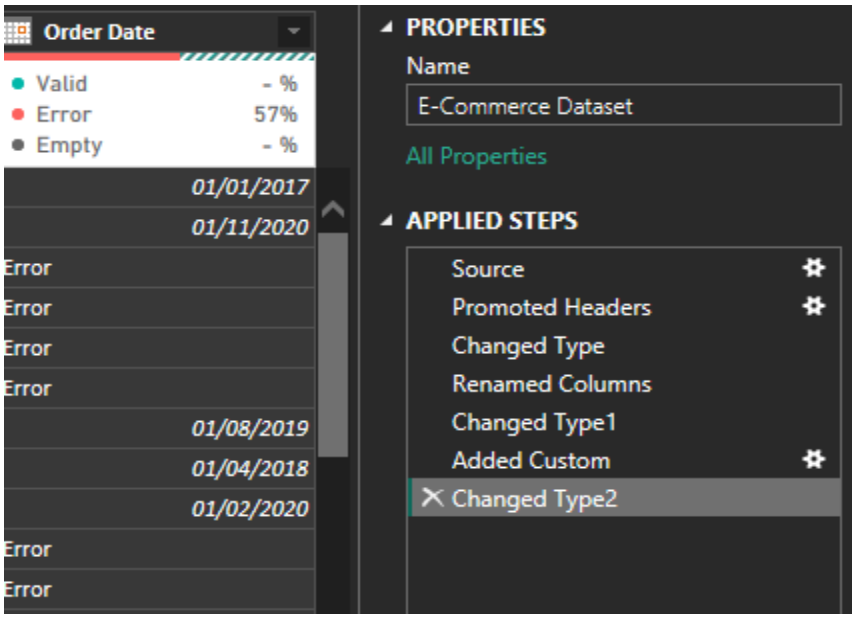


Figure 26 Data type changed to Date/Time

- Errors were removed from the Order Date column.

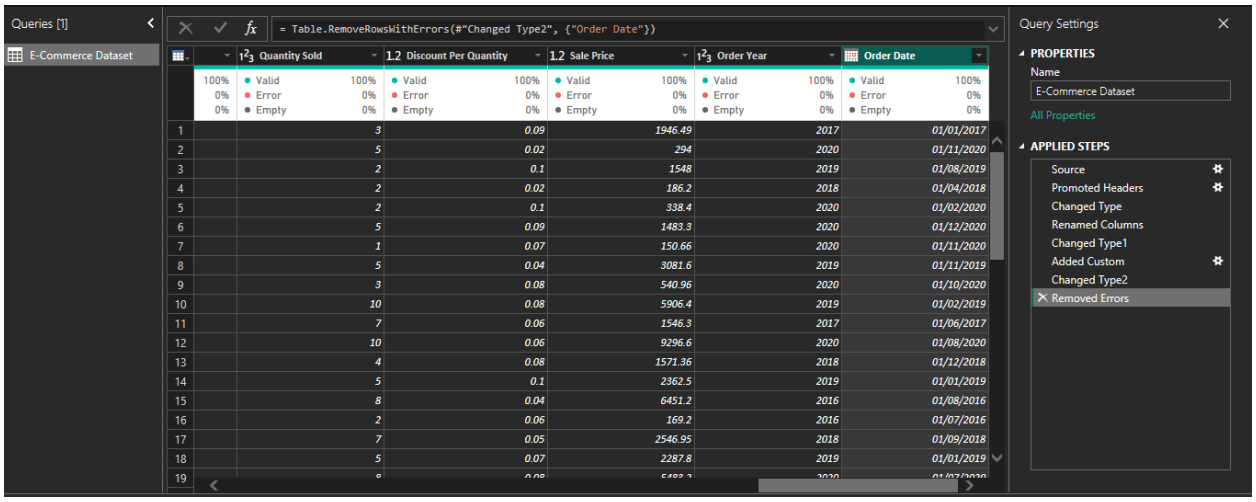


Figure 27 Final Order Date column

- After creation, the **Profit** column data type was explicitly converted to **Decimal Number**.

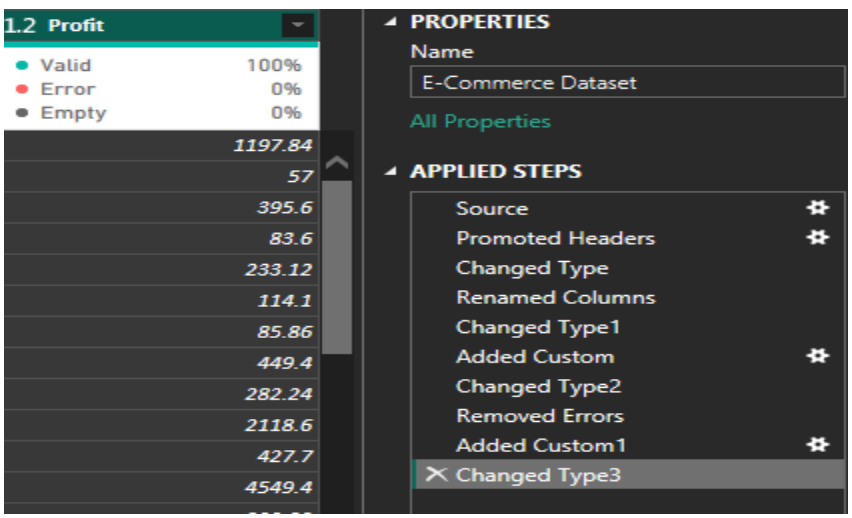


Figure 28 Profit Data Type Standardization

5. Data Validation and Quality Checks

- Column profiling was used to verify:
  - **100% valid values** for key analytical fields
  - No unexpected nulls or errors in critical columns such as Sales, Quantity, and Customer attributes
- Minor inconsistencies were resolved during transformation.

- The dataset contained no duplicate rows requiring removal.

	Customer ID	Customer Name	Customer Email	Customer Gender	Customer IP	Pro
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Vali Erro Emp
1	4070	Roberta Lesch	Rodrick_Kassulke88@gmail.com	Female	138.152.90.54	Hat
2	2530	Adaline Hoeger	Vicky93@yahoo.com	Female	35.240.213.166	Chee
3	2697	Yesenia Hahn	Linda48@gmail.com	Male	180.226.79.181	Chip
4	2405	Fern Stark	Eriberto5@yahoo.com	Female	227.148.70.138	Chai
5	4394	Ronaldo Bashirian	Nicole56@hotmail.com	Male	85.197.31.229	Chic
6	2013	Moses Runolfsdottir	Polly99@hotmail.com	Male	118.177.129.243	Sala
7	396	Jared Purdy	Nicholaus77@hotmail.com	Female	6.136.10.26	Keyb
8	4151	Minerva O'Kon	Daphne_Kessler@yahoo.com	Female	122.246.199.255	Glov
9	2987	Shana Beier	Valentine89@gmail.com	Female	129.204.15.31	Pant
10	26	Tristin Bogan	Jaylen15@hotmail.com	Female	69.224.113.170	Keyb

Figure 29 Data Validation and Quality Checks-I

	Product Name	Product Price	Cost Of Goods	Product Category	Country	
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	V E E
1	Hat	713	249.55	Movies	Seychelles	A
2	Cheese	60	47.4	Health	Qatar	A
3	Chips	310	145.7	Sports	Cameroon	A
4	Chair	468	149.76	Beauty	Guadeloupe	N
5	Chicken	360	140.4	Health	Swaziland	A
6	Salad	190	119.7	Health	Costa Rica	N
7	Keyboard	860	576.2	Health	Kuwait	A
8	Gloves	95	51.3	Music	Kazakhstan	A
9	Pants	188	52.64	Toys	Andorra	E
10	Keyboard	915	356.85	Baby	Tunisia	A

Figure 30 Data Validation and Quality Checks-II

	Continent	Order Time	Order Day	Order Month	Quantity Sold	Discount Pe
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	Africa	1:43:00 pm	1	February	3	
2	Asia	12:25:00 pm	11	February	5	
3	Africa	8:14:00 pm	25	September	2	
4	North America	1:13:00 pm	30	June	1	
5	Africa	10:50:00 pm	28	September	6	
6	North America	11:15:00 pm	30	December	5	
7	Asia	5:56:00 pm	8	June	2	
8	Asia	9:34:00 pm	4	May	2	
9	Europe	8:15:00 pm	2	April	2	
10	Africa	10:57:00 pm	24	June	1	

Figure 31 Data Validation and Quality Checks-III

		Order Month	Quantity Sold	Discount Per Quantity	Sale Price	Order Year
	100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	1	February	3	0.09	1946.49	2017
2	11	February	5	0.02	294	2020
3	25	September	2	0.03	601.4	2020
4	30	June	1	0.1	421.2	2020
5	28	September	6	0.08	1987.2	2020
6	30	December	5	0.09	864.5	2020
7	8	June	2	0.1	1548	2019
8	4	May	2	0.02	186.2	2018
9	2	April	2	0.1	338.4	2020
10	24	June	1	0.01	905.85	2020

Figure 32 Data Validation and Quality Checks-IV

6. Handling of Null Values

During data validation, **null values were identified in two numeric columns** that are critical for sales and discount analysis.

a. Discount Per Quantity

- The **Discount Per Quantity** column contained null values for transactions where no discount was applied.
- These nulls were **replaced with 0** using the *Replace Values* function in Power Query.
- This approach was chosen because a missing discount logically represents **no discount applied**, rather than missing data.

Discount Per Quantity
Valid 90%
Error 0%
Empty 10%
0.09
0.02
null
0.02
null
0.09
0.07
0.04
0.08
0.08
0.06
0.06
0.08
null

Figure 33 Null values found in Discount per quantity column

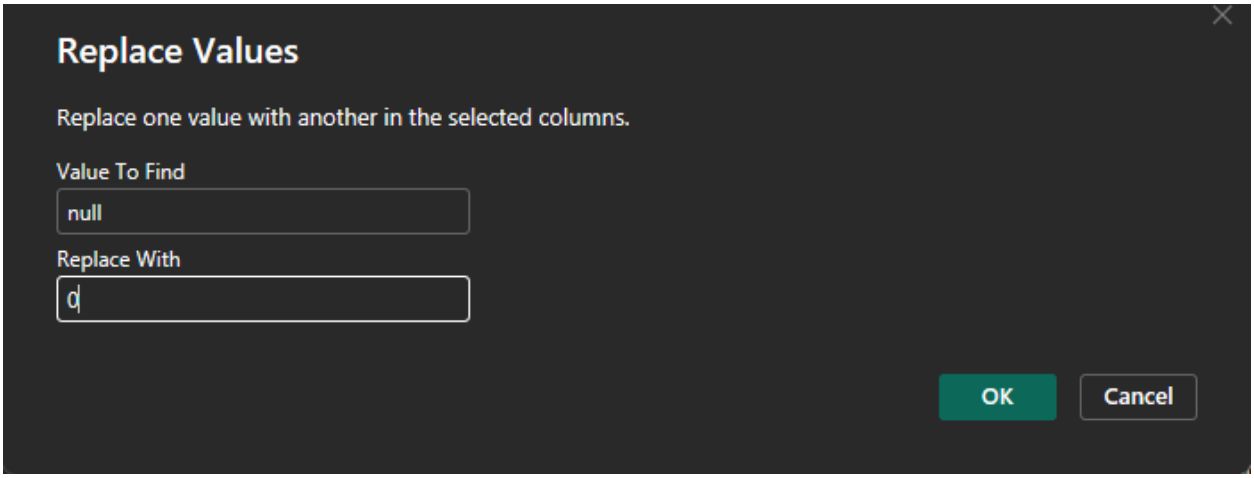


Figure 34 Replaced null values with zero in Discount per quantity column

b. Product Price Column

- A second numeric column (sales-related) also contained null values.
- These values were reviewed and **handled appropriately to prevent calculation errors**, ensuring consistency in aggregations.

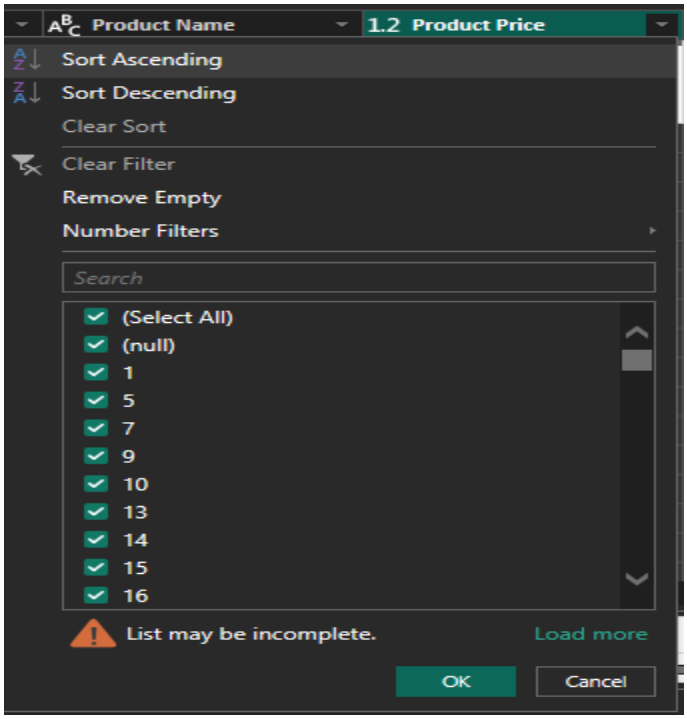


Figure 35 Found Null values in Product Price

- Replacing nulls avoided issues in measures such as **total sales, average order value, and profit calculations**.

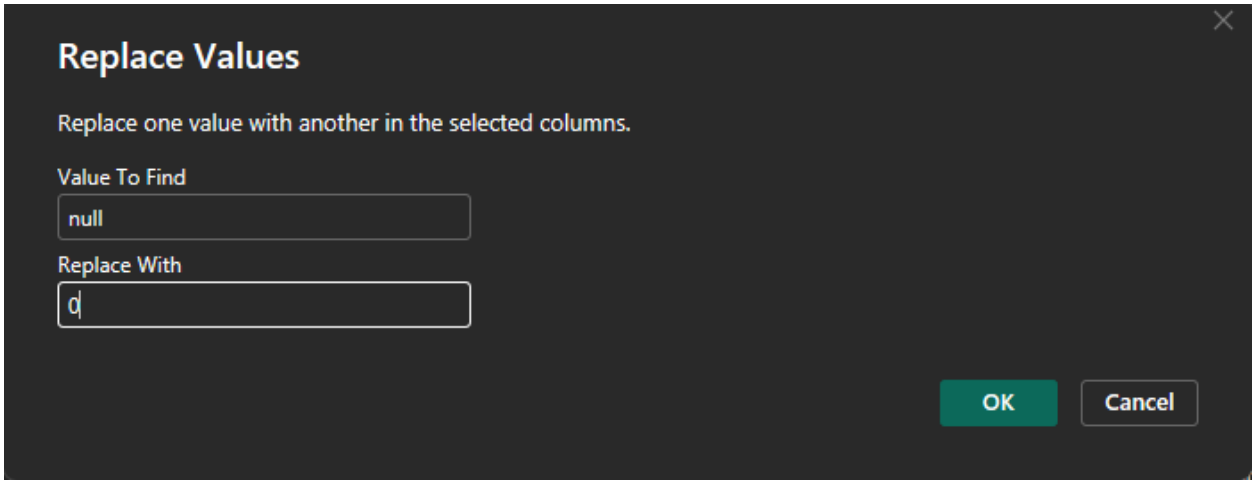


Figure 36 Replaced null values with zero in Product Price column

7. Conditional Formatting (Total Sales)

- **Conditional background colour formatting** was applied to the **Total Sales** field using a **gradient scale**.
- Colours range from **lighter to darker shades**, representing lower to higher sales values.
- Formatting is applied to **values only**, with empty values treated as zero.

Purpose:

- Provides immediate **visual emphasis on high and low sales values**
- Enables faster comparison and pattern recognition in tabular visuals
- Enhances dashboard readability for business users
- Supports intuitive, insight-driven decision making without relying solely on numeric values

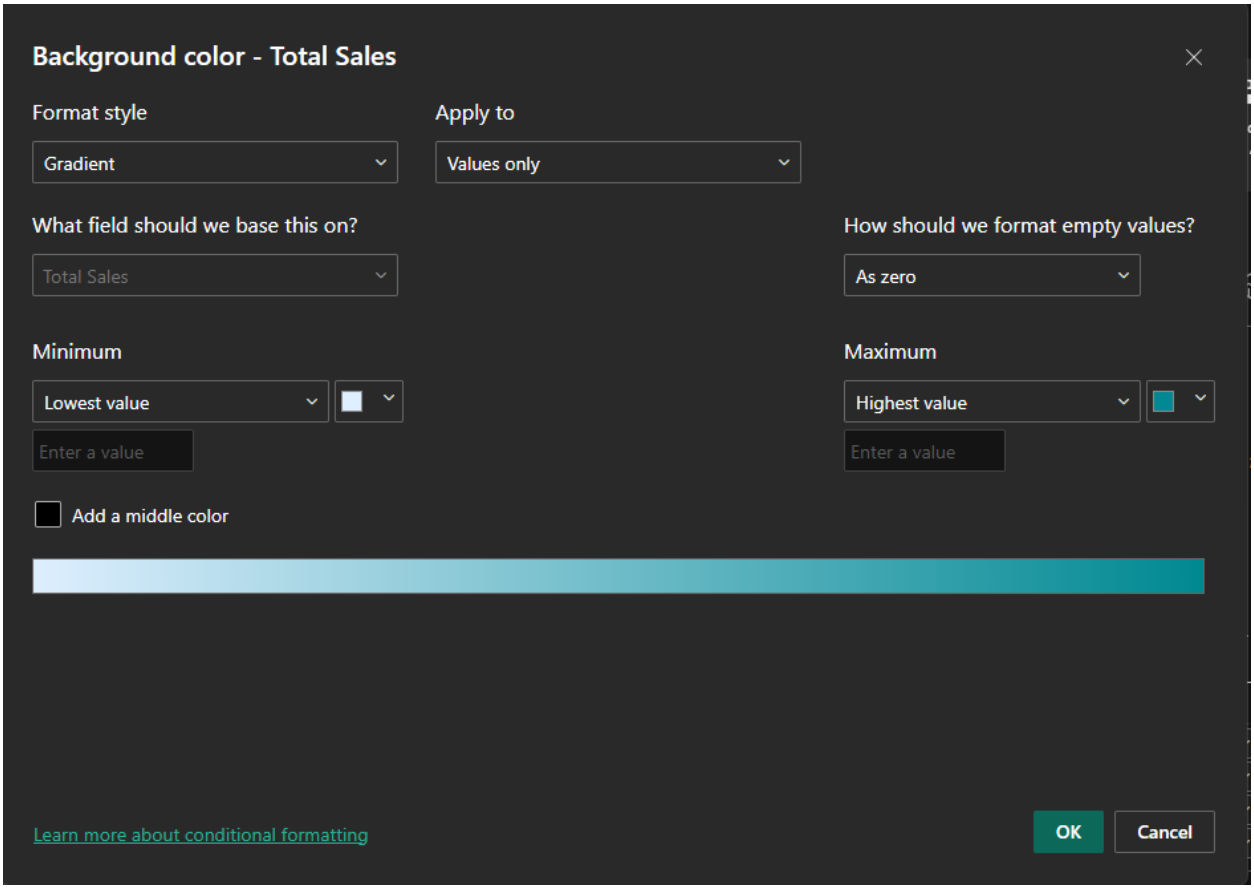


Figure 37 Conditional formatting on Total Sales

viii. Product Price Binning

- A **bin group** was created on the **Product Price** field to divide continuous price values into **equally sized intervals**.
- The binning was defined using a **fixed bin size**, covering the full product price range.
- This transformation converts a continuous numeric variable into **discrete price ranges**.

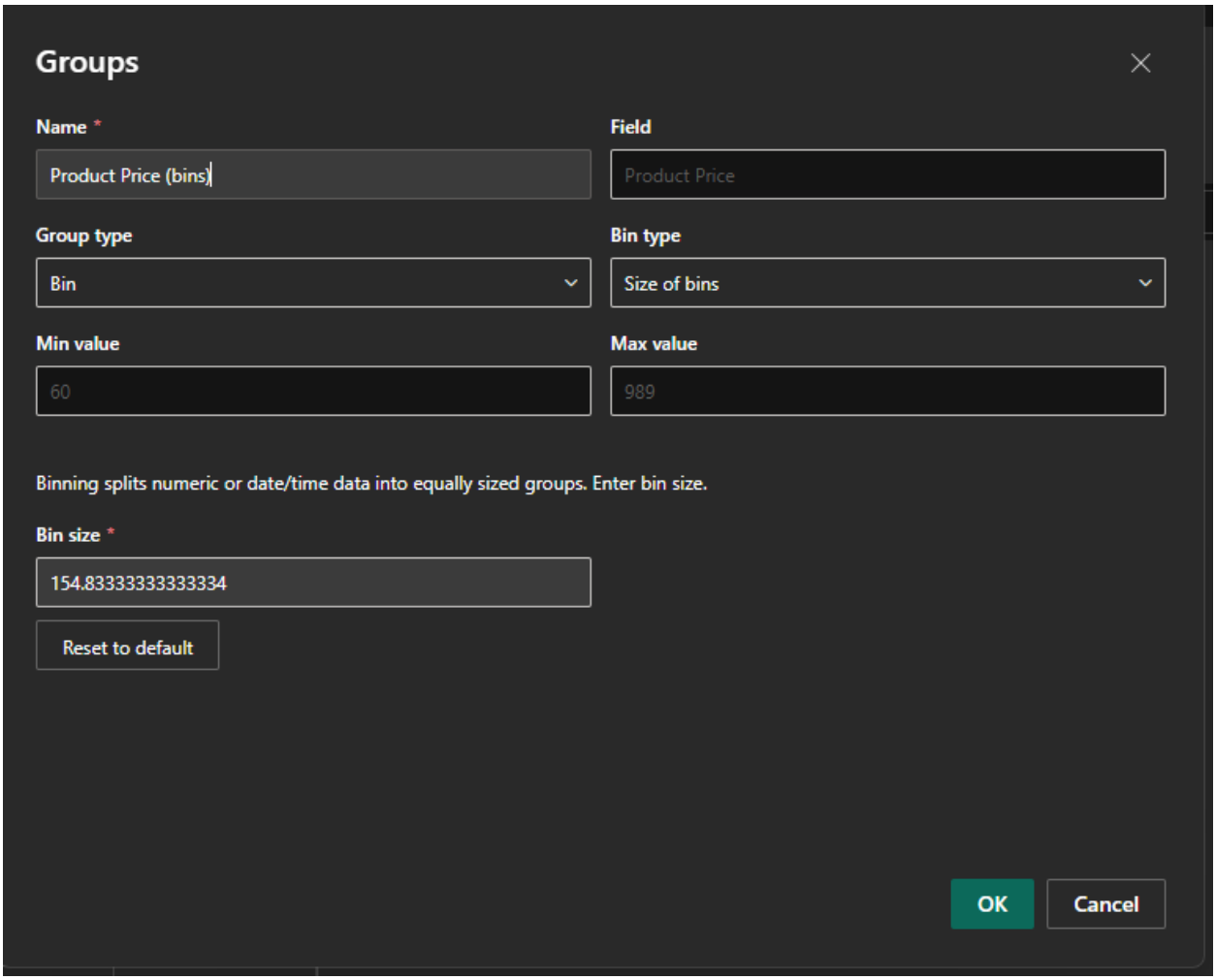


Figure 38 Product Price Binning

ix. Creation of Conditional Column – Order Value Category

- A **conditional column** named **Order Value Category** was added to the **FactSales** table to classify transactions based on their **Sale Price**.
- The classification rules applied were:
  - **High**: Sale Price greater than 1500
  - **Medium**: Sale Price less than 1500
  - **Low**: Sale Price less than 500
- This transformation converts a continuous numeric variable into **meaningful categorical segments**.

Each transaction is now labelled as **High**, **Medium**, or **Low** value, enabling segmented analysis.

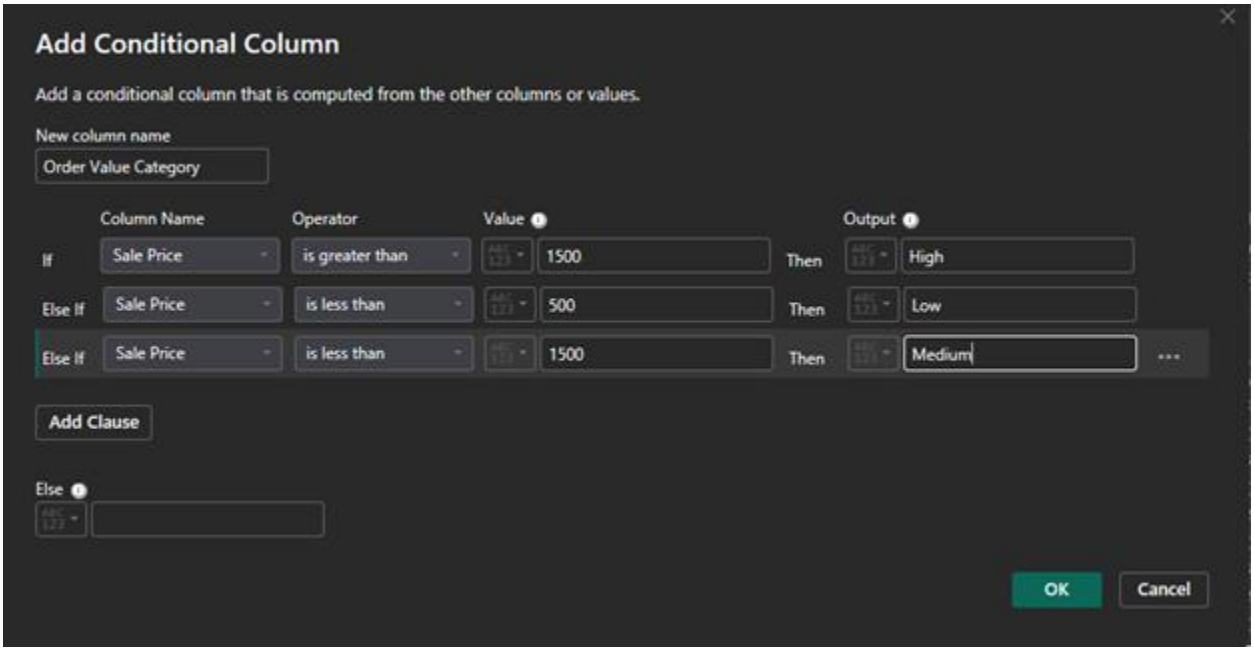


Figure 39 Conditional Column

x. Creation of Dimension Tables and Fact Table

1. Fact Table Creation - FactSales

- A central **FactSales** table was created from the cleaned E-Commerce dataset.
- Non-analytical and redundant columns (e.g., customer contact details) were removed to reduce data redundancy.
- The fact table retains only **transaction-level measures and foreign keys**, including:
    - Customer ID
    - Product Name
    - Country
    - Quantity Sold
    - Sale Price
    - Cost of Goods

- Profit

Purpose:

To store measurable business events such as sales, cost, quantity, and profit at the lowest granularity.

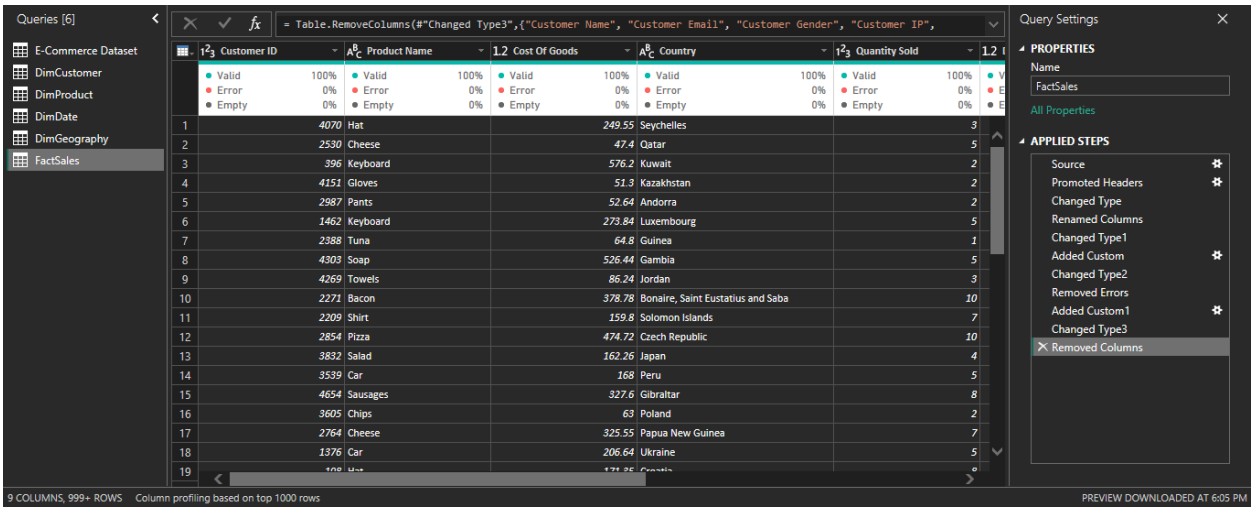


Figure 40 Fact Table

2. Customer Dimension Table – DimCustomer

- A separate **DimCustomer** table was created by:
  - Selecting customer-related columns from the source dataset
  - Removing duplicates using Customer ID
  - Removing non-essential fields such as Customer Email and Customer IP
- Final attributes include:
  - Customer ID
  - Customer Name
  - Customer Gender

Purpose:

To enable customer-level analysis (e.g., revenue by customer, gender-based insights) while avoiding duplication in the fact table.

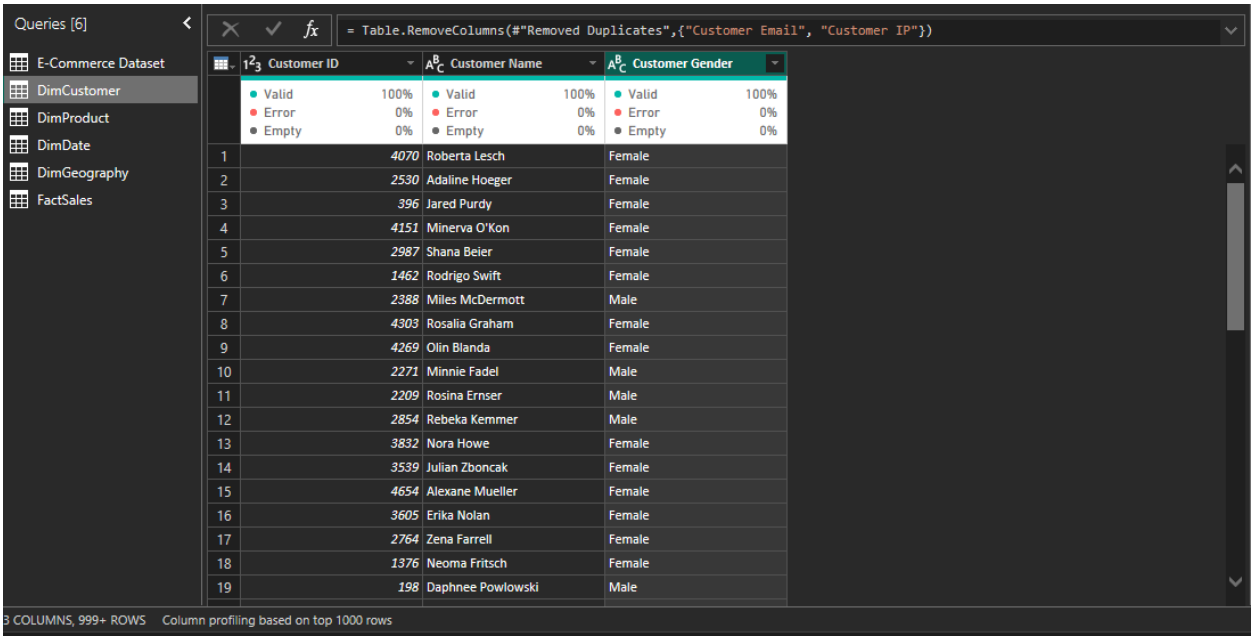


Figure 41 Customer Dimension Table

3. Product Dimension Table– DimProduct

- The **DimProduct** table was created by:
  - Extracting product-related attributes
  - Removing duplicate records using Product Name
- Final attributes include:
  - Product Name
  - Product Price
  - Cost of Goods
  - Product Category

Purpose:

To support product and category-level analysis such as profitability, contribution, and ranking.

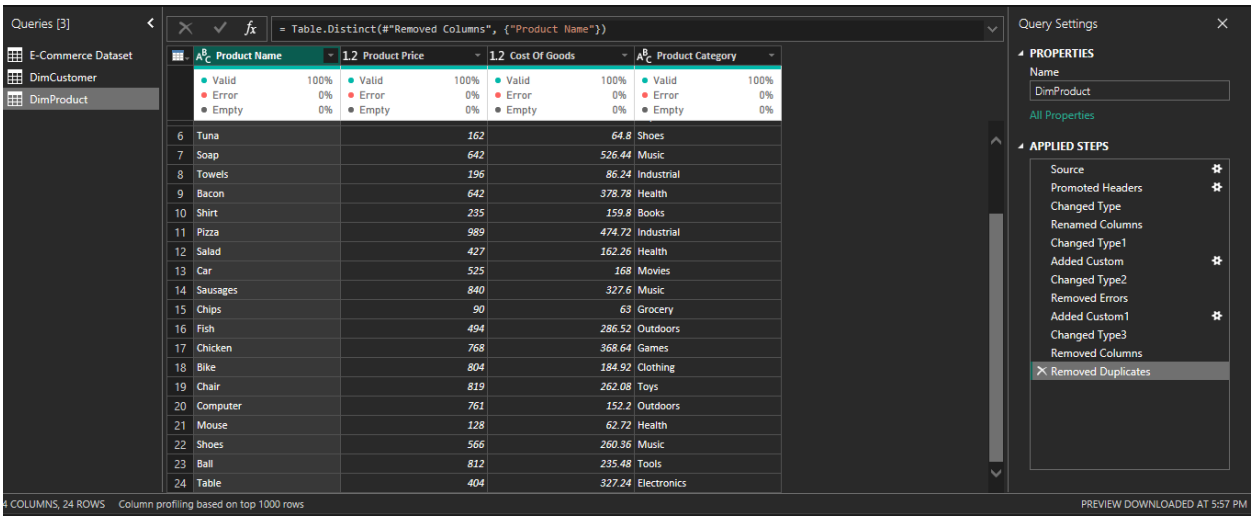


Figure 42 Product Dimension Table

4. Date Dimension Table – DimDate

- A **DimDate** table was created using:
  - Order Day
  - Order Month
  - Order Year
  - Custom Order Date
- Duplicate dates were removed to ensure uniqueness.

Purpose:

To enable time-based analysis such as yearly trends, monthly patterns, and seasonality using a dedicated date table.

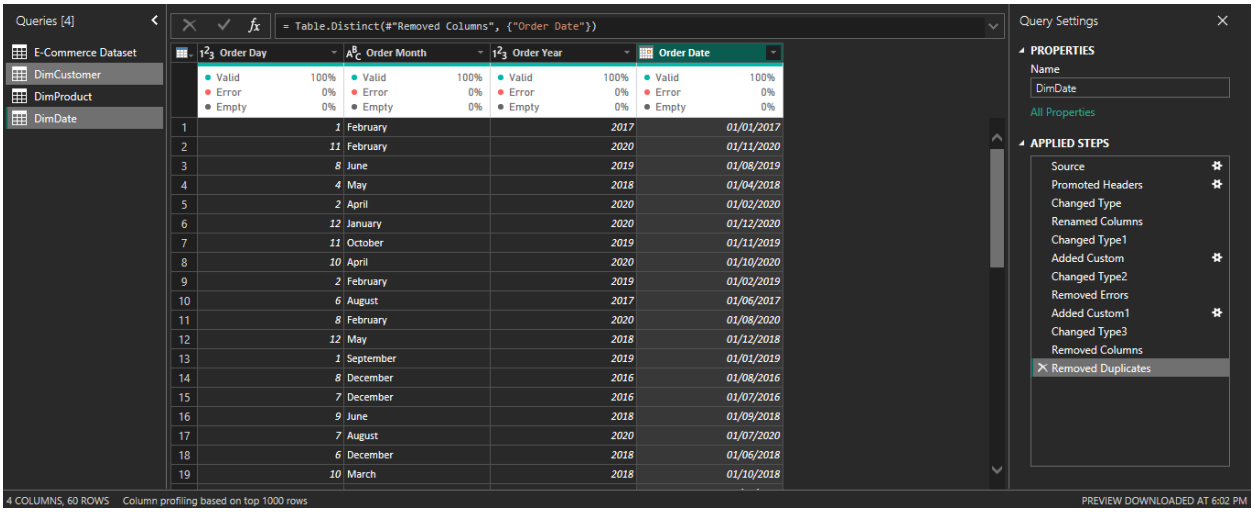


Figure 43 Date Dimension Table

5. Geography Dimension Table – DimGeography

- The **DimGeography** table was created by:
  - Extracting Country and Continent columns
  - Removing duplicate country records
- Final attributes include:
  - Country
  - Continent

Purpose:

To support geographic analysis and regional aggregation without repeating location data in the fact table.

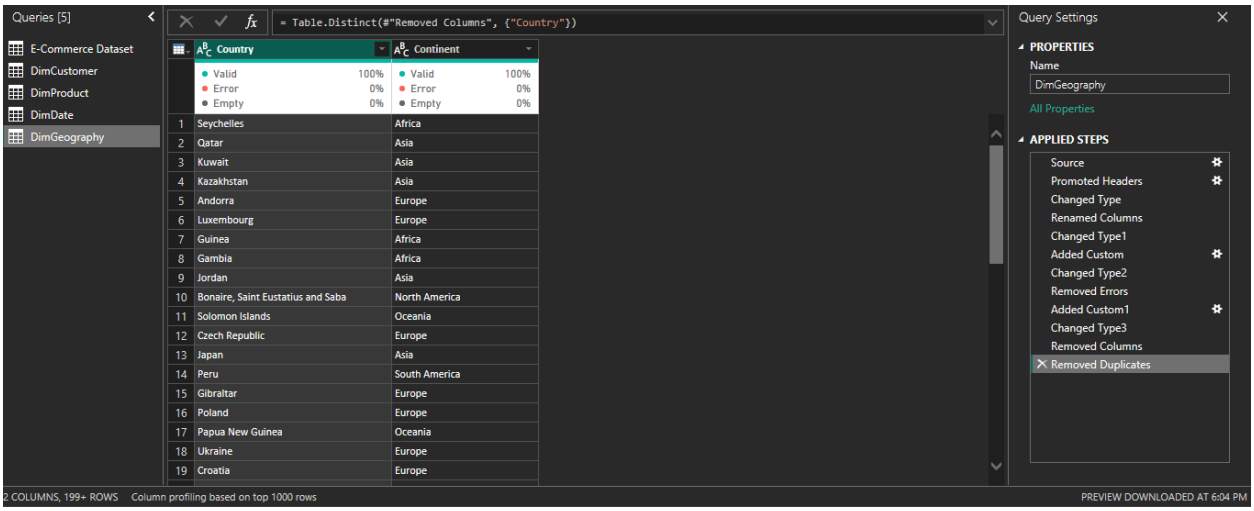


Figure 44 Geography Dimension Table

3.2 BI Data Modelling via Star Schema – Facts and Dimensions

A star schema data model was implemented to support efficient querying, accurate aggregation, and scalable business intelligence reporting. This approach separates transactional data (facts) from descriptive attributes (dimensions), ensuring clarity and performance.

Before Star Schema Implementation

Before applying the star schema, the data model relied heavily on the raw E-Commerce Dataset, which contained customer, product, date, geography, and sales attributes in a single table.

This resulted in:

- Data redundancy across attributes
- Mixed transactional and descriptive data
- Higher risk of ambiguous relationships and incorrect aggregations
- Limited scalability and reduced model clarity

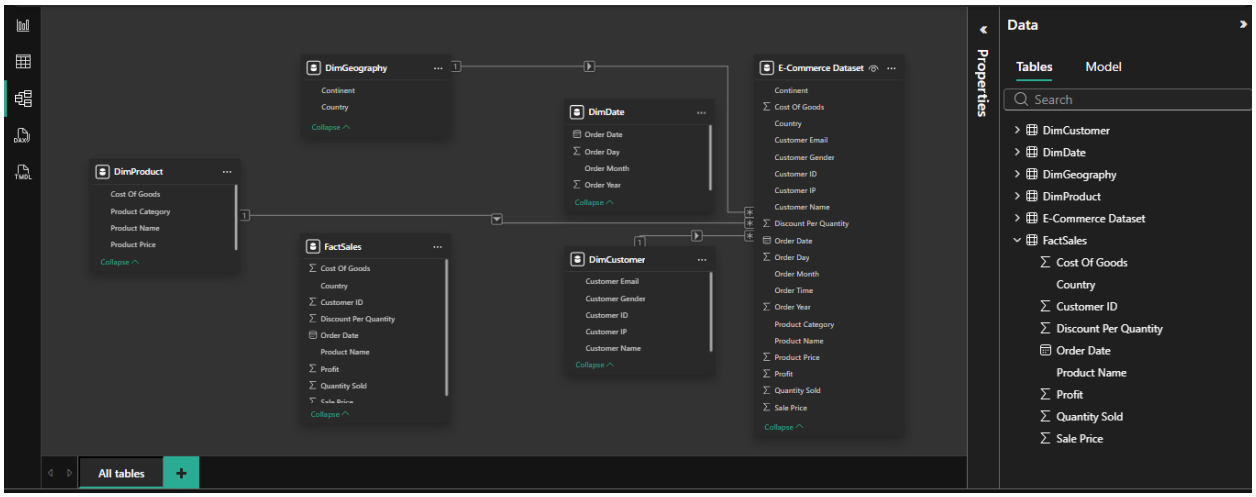


Figure 45 Before Star Schema

After Star Schema Implementation

After validation, the model was restructured into a clean star schema:

- FactSales acts as the central transactional table
- Supporting dimension tables include:
    - DimCustomer
    - DimProduct
    - DimDate
    - DimGeography
- The raw E-Commerce Dataset table was retained only as a staging/reference table and removed from analytical relationships

All relationships use:

- One-to-many cardinality (Dimension → Fact)
- Single-directional filtering (best practice)

Final Relationship Design

Dimension Table	Join Key	Relationship
DimCustomer	Customer ID	One-to-Many
DimProduct	Product Name	One-to-Many
DimDate	Order Date	One-to-Many
DimGeography	Country	One-to-Many

Benefits of the Optimised Model

- Eliminates data redundancy
- Improves query performance
- Prevents ambiguous relationships
- Simplifies DAX calculations
- Aligns with industry-standard BI modelling practices

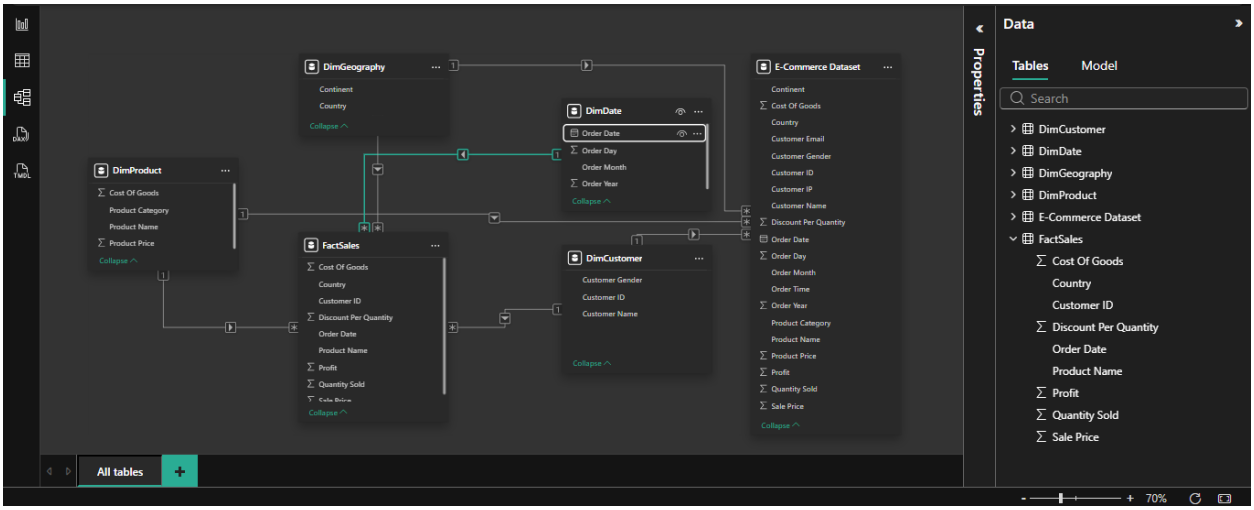


Figure 46 After Star Schema

3.3 DAX and M Language

In this report, **DAX measures** were created to calculate core KPIs and support ranking, profitability, and contribution analysis across the star schema model. These measures enable dynamic aggregation that responds to slicers (e.g., Year, Location, Category) and cross-filtering between visuals.

i. **Total Sales**

Used to measure overall revenue generated and to analyse sales performance across products, customers, time periods, and regions.

```
Total Sales = SUM ( FactSales[Sale Price] )
```

Figure 47 Total Sales

ii. **Total Profit**

Used to evaluate overall business profitability and support profit-based performance analysis.

```
Total Profit = SUM ( FactSales[Profit] )
```

Figure 48 Total Profit

iii. **Total Quantity Sold**

Used to track sales volume and identify high-demand products and categories.

```
Total Quantity Sold = SUM ( FactSales[Quantity Sold] )
```

Figure 49 Total Quantity Sold

iv. **Profit Margin %**

Used to assess profitability efficiency by comparing profit relative to sales revenue.

```
Profit Margin % = DIVIDE ( [Total Profit], [Total Sales], 0 )
```

Figure 50 Profit Margin %

v. **Average Order Value**

Used to understand the average revenue generated per order and evaluate customer spending behaviour.

```
Average Order Value = DIVIDE ( [Total Sales], DISTINCTCOUNT ( FactSales[Order Date] ), 0 )
```

Figure 51 Average Order Value

vi. **Loss Orders**

Used to identify and count transactions that resulted in negative profit, supporting risk and margin analysis.

```
Loss Orders = CALCULATE (COUNTROWS ( FactSales ),FactSales[Profit] < 0)
```

Figure 52 Loss Orders

vii. **Customer Rank**

Used to rank customers based on revenue contribution and identify top-performing customers.

```
Customer Rank =
RANKX (
    ALL ( DimCustomer[Customer Name] ),
    [Total Sales],
    ,
    DESC
)
```

Figure 53 Customer Rank

viii. **Average Discount**

Used to analyse discounting behaviour and assess the impact of discounts on pricing and profitability.

```
Average Discount =AVERAGE ( FactSales[Discount Per Quantity] )
```

Figure 54 Average Discount

ix. **Discounted Orders**

Used to count orders where discounts were applied, supporting promotion and pricing strategy analysis.

```
Discounted Orders =
CALCULATE (
    [Total Orders],
    FactSales[Discount Per Quantity] > 0
)
```

Figure 55 Discounted Orders

x. **Cumulative Sales %**

Used to perform Pareto (80/20) analysis by identifying products that contribute the majority of total sales.

```
Cumulative Sales % =
VAR TotalSalesAllProducts =CALCULATE ( [Total Sales], ALL ( DimProduct[Product Name] ) )
VAR CurrentSales =[Total Sales]
VAR CumSales =
    CALCULATE (
        [Total Sales],
        FILTER (
            ALL ( DimProduct[Product Name] ),
            [Total Sales] >= CurrentSales
        )
    )
RETURN
DIVIDE ( CumSales, TotalSalesAllProducts, 0 )
```

Figure 56 Cumulative Sales %

xi. Creation of Calculated Columns

a) Order Date

- A custom **Order Date** column was created using Power Query:
    - Derived from Order Year and Order Day
    - Implemented using the formula: = #date([Order Year], [Order Day], 1)
- This enabled time-based analysis such as **yearly trends, monthly patterns, and seasonality**.

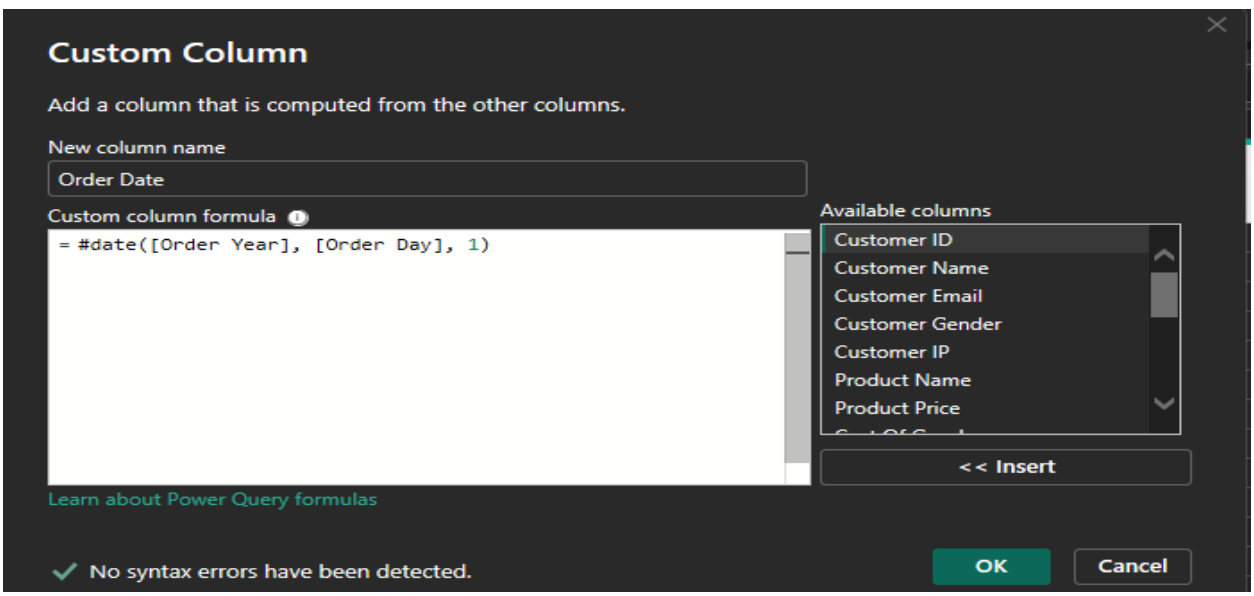


Figure 57 Creation of Custom Date Column

b) Profit Column

- A new calculated column **Profit** was created in **Power Query** to represent transactional profitability.
- The column was derived using the following formula:
- `Profit = [Sale Price] - ([Cost Of Goods] * [Quantity Sold])`
- This calculation ensures that profit reflects the **true cost-adjusted value per transaction**, rather than relying on product price alone.
- The formula was validated in Power Query, confirming **no syntax errors** and **100% valid values** across rows.

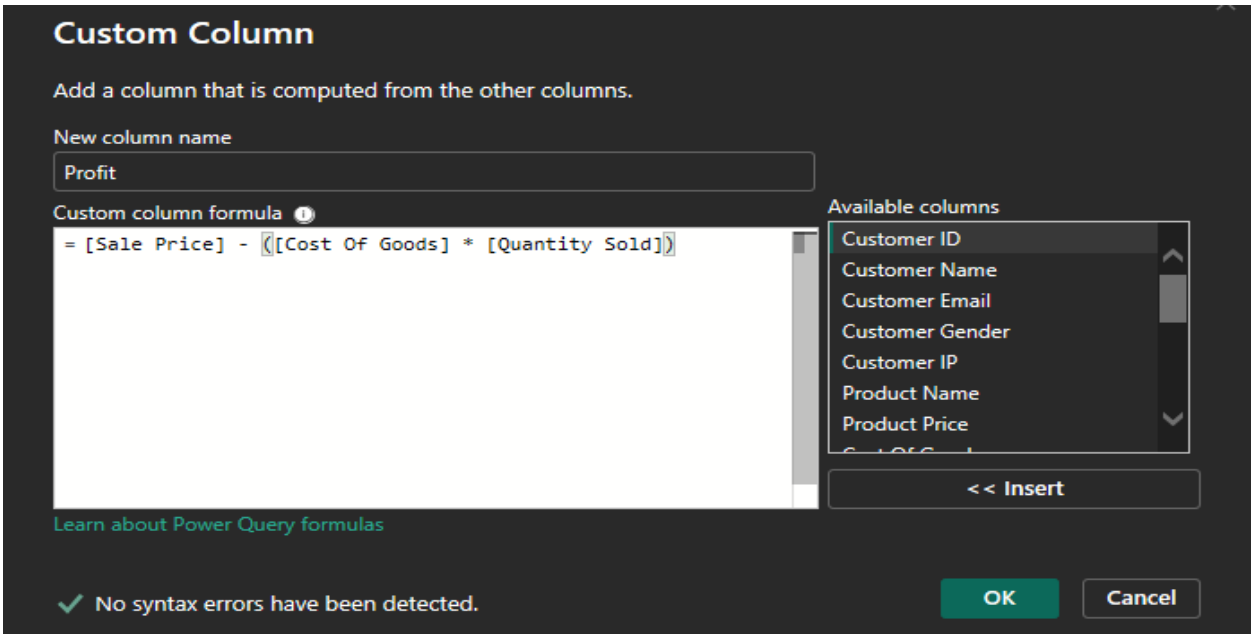


Figure 58 Profit Column creation

c) Profit Tier

The Profit Tier calculated column categorizes transactions into low, medium, and high profit groups based on their deviation from average profit, enabling clearer profitability analysis.

```
Profit Tier =
VAR AvgProfit =
    AVERAGEX ( ALL ( FactSales ), FactSales[Profit] )
RETURN
SWITCH (
    TRUE(),
    FactSales[Profit] < AvgProfit * 0.75, "Low Profit",
    FactSales[Profit] < AvgProfit * 1.25, "Medium Profit",
    "High Profit"
)
```

Figure 59 Profit Tier

3.4 Dashboard

The Power BI dashboard is designed as a **multi-page, interactive analytical report** that provides both **high-level executive insights** and **detailed drill-down analysis**. A consistent layout, colour theme, and navigation panel are used across all pages to ensure usability and coherence.

Dashboard Structure

- A **left-side navigation menu** allows users to move between pages: *Home, Executive Overview, Sales Trends, Products, Customers, and Sales Drivers*.
- **Global slicers** (Year, Location, Category) are placed consistently on each page to enable cross-page filtering and interactive analysis.
- Each page focuses on a **specific analytical theme**, reducing visual clutter and improving insight clarity.

Page Organisation

i. *Home Page*

Serves as the landing page, presenting the dashboard title and guiding users into the analysis sections.

Figure 60 Home Page

ii. *Executive Overview*

Displays key KPIs such as Total Sales, Total Orders, Net Profit, Profit Margin, and Average Order Value. Supporting visuals summarise revenue trends, profit contribution by category, regional sales, and performance against targets.

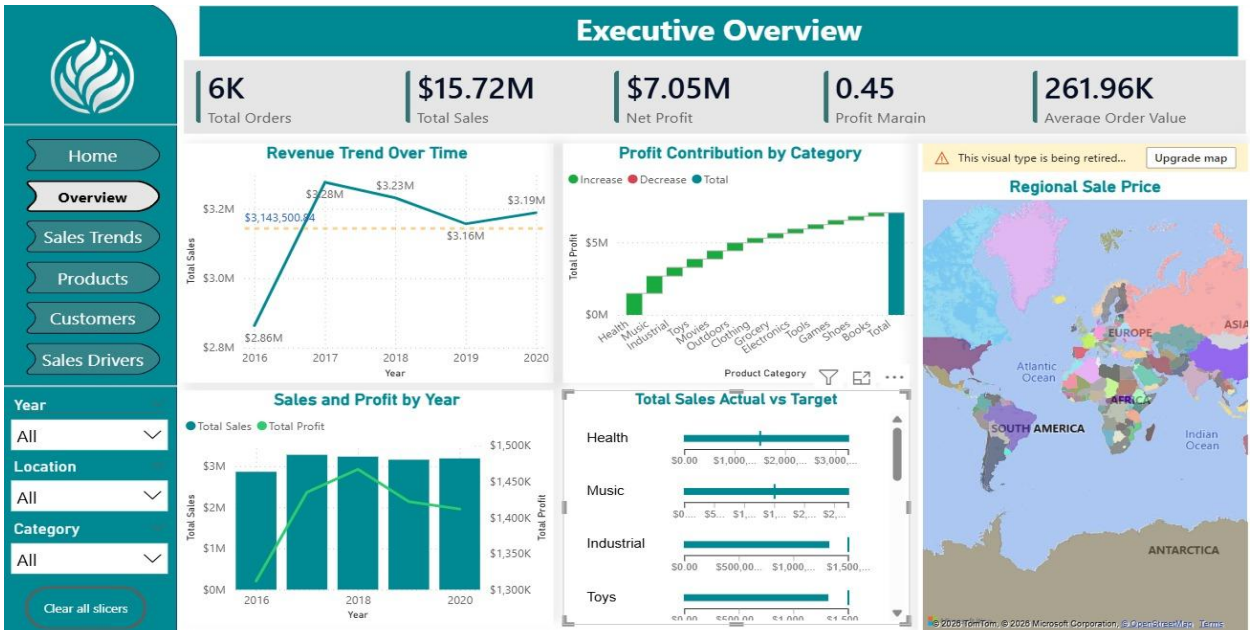


Figure 61 Executive Overview

iii. *Sales Trends Analysis Page*

Focuses on temporal analysis, including yearly and monthly sales patterns, sales composition trends, profit margin performance, and changing category sales share over time.

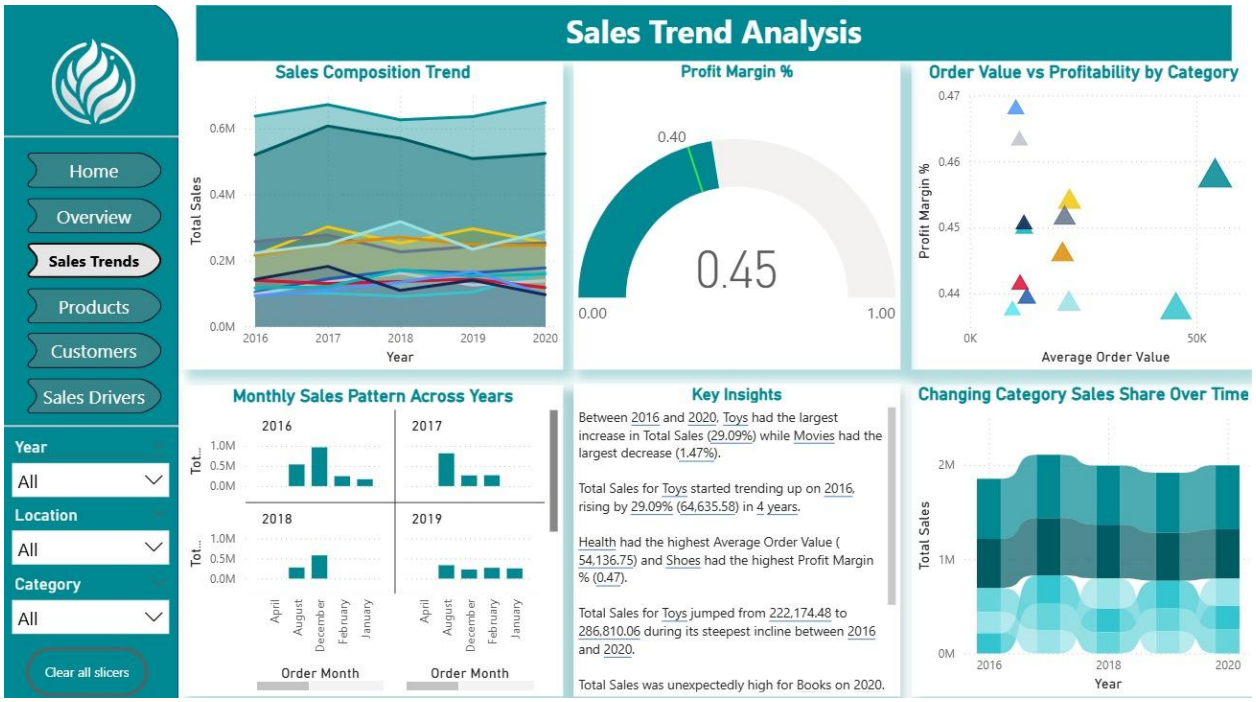


Figure 62 Sales Trends Analysis Page

iv. *Products Analysis Page*

Provides product-level insights such as sales and profit by category, price distribution, Pareto (80/20) contribution analysis, and product profitability positioning.

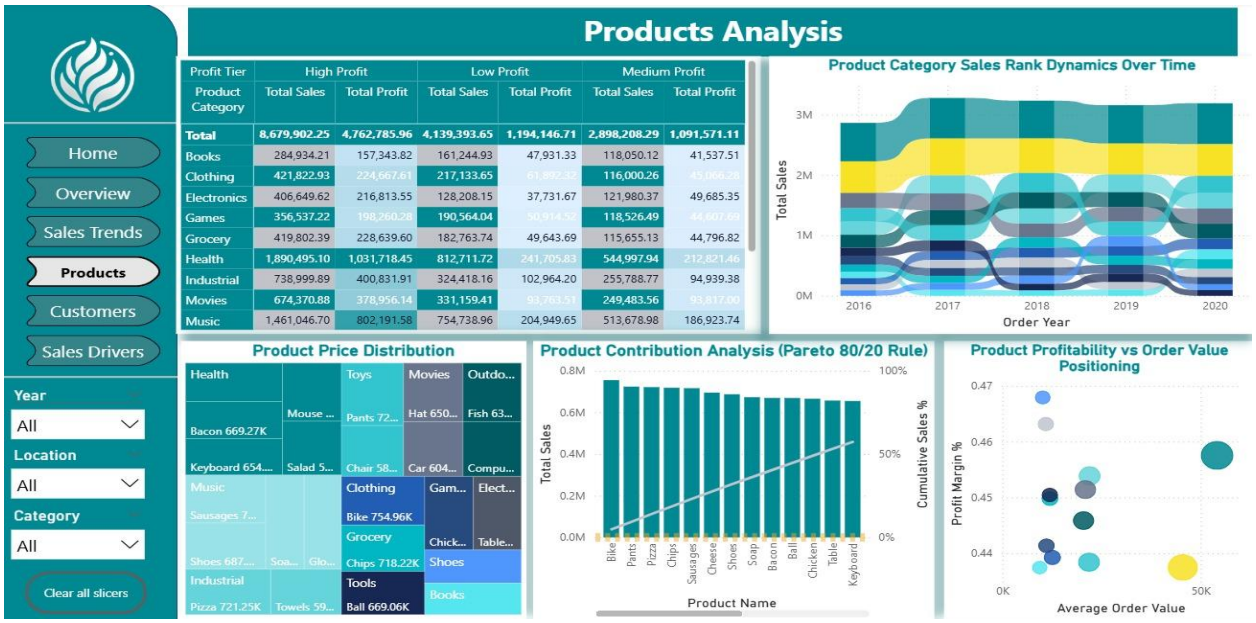


Figure 63 Products Analysis Page

v. *Customer Analysis Page*

Analyses customer behaviour through spend distribution, value segmentation, top revenue contributors, revenue contribution by gender, and average order value distribution.

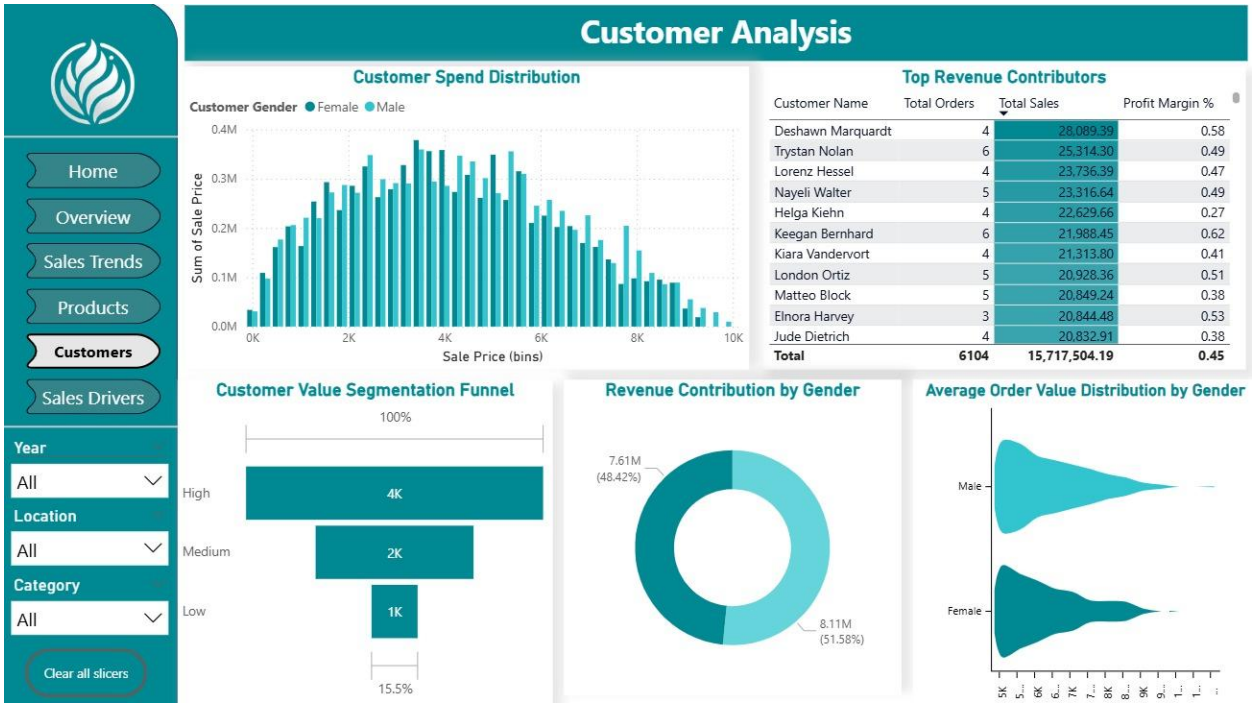


Figure 64 Customer Analysis Page

vi. *Sales Drivers Analysis Page*

Uses a decomposition-style visual to identify key drivers of total sales by product category, geography, and customer gender, supporting root-cause analysis.

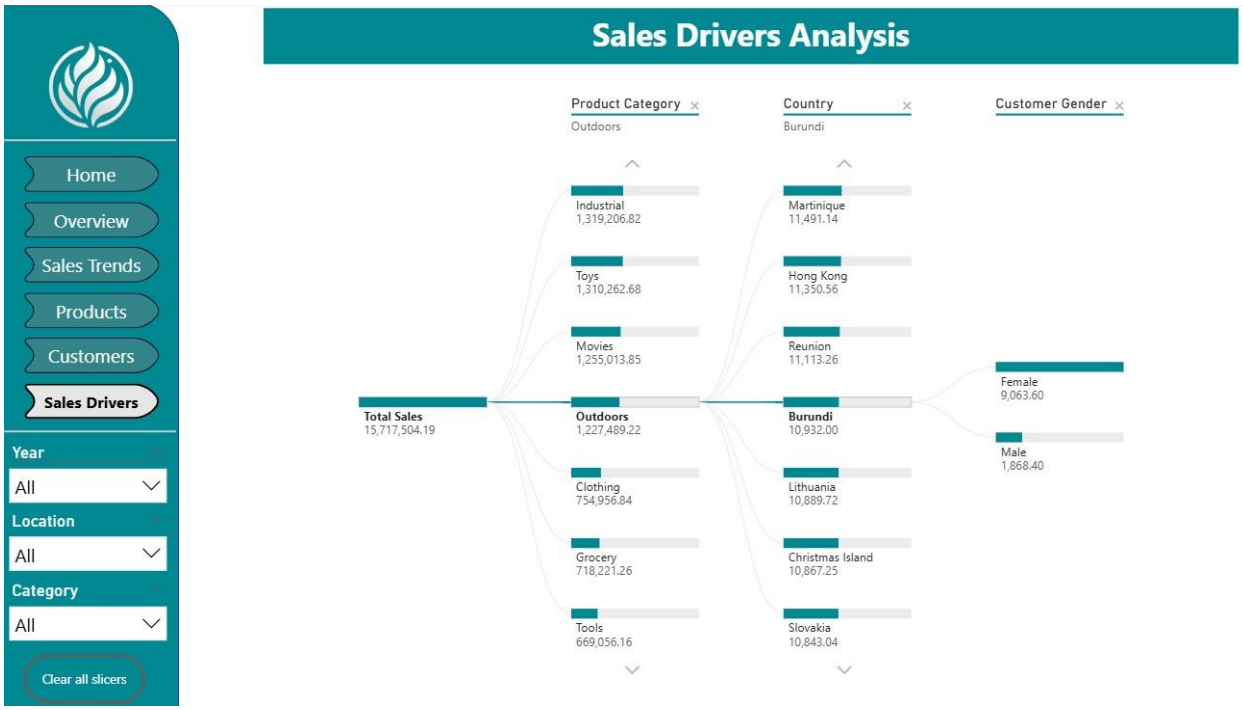


Figure 65 Sales Drivers Analysis Page

The dashboard is organised using a **thematic, page-based structure**, progressing from **strategic overview to detailed analysis**. Consistent slicers, navigation, and visual design ensure an intuitive user experience while enabling comprehensive business insight discovery.